Modern Force and Coercive Backfire

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Abstract. Modern military force enables efficient operations short of general war through automation and AI-assisted precision strikes. Using a parsimonious bargaining framework, we show that low-cost military technology can raise the political cost of conflict. Coercion requires both threat and assurance: resistance must be costly, and compliance must prevent further harm. Existing theory treats the two as an inverse trade-off. A subwar operation can instead weaken both. By replacing war after resistance, it reduces punishment; by replacing settlement after compliance, it reduces protection. These losses jointly reduce the target's equilibrium concession, and the smaller concession further weakens assurance by making war more attractive after compliance. This response produces three implications. It can make every coercer type worse off despite adding an option. It can induce types that otherwise settle to choose general war. It can also make cheaper operations raise total conflict cost. Operational efficiency can therefore make coercion more destructive.

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Introduction

Interstate wars have become less frequent (Pinker 2011; Altman 2020). Whether violence itself has declined remains contested (Braumoeller 2019; Altman 2020), yet both sides of this debate count wars. That focus is increasingly incomplete. Modern military force gives states efficient ways to use force short of general war. AI-assisted operations, for example, combine armed drones with persistent surveillance and precision strike while limiting personnel exposure (Zegart 2018; Horowitz 2020). States can seize positions incrementally, disrupt production through cyber operations, or launch stand-off strikes while trying to remain below the threshold of general war (Schram 2021; Gannon et al. 2024; Kenkel and Schram 2025). Such operations can replace general war with more limited force, but they also make violence feasible where war would be too costly. Fewer wars therefore need not mean less violence, and violence remains common in international politics (Davies, Pettersson, and Öberg 2022). These direct effects do not determine whether the capability promotes peace, because it can also change the bargain struck before force is used. How does acquiring a subwar capability change the concession a target is willing to make, and does that capability broaden or narrow the prospects for peace?

Existing theory treats threat and assurance as an inverse trade-off: measures that make resistance more costly also make post-compliance restraint less credible, while measures that strengthen restraint weaken the threat (Schelling 1966; Kydd and McManus 2017; Cebul, Dafoe, and Monteiro 2021; Pauly 2024). A subwar capability can instead weaken both. When the operation replaces a more damaging war after resistance, resistance becomes less costly and the threat weakens. The target then accepts a smaller demand. Because a smaller transfer makes general war more attractive to the coercer after compliance, the demand response increases post-compliance harm and weakens assurance. The operation can also strengthen either component by replacing settlement after resistance or war after compliance; the joint loss arises under the conditions derived below. We call the resulting bargaining performance *coercive ability*: the expected portion of an accepted demand that remains in the coercer's hands without general war. The demand is the price the target agrees to pay; coercive ability discounts that amount by the probability that general war later reopens the dispute.

We study the subwar capability in a parsimonious bargaining framework of interstate conflict built from two minimal extensions to a canonical crisis-bargaining model. The canonical model has an informed coercer choose a demand and the target comply or resist (Banks 1990; Fearon 1994; Schultz 1999). First, we keep the coercer's continuation choice open after both resistance and compliance. This allows us to study coercion as a dual problem: making resistance costly while assuring the target that compliance will prevent further harm. Second, we add a subwar operation that costs less than war to the same continuation menu. The benchmark incorporates the first extension; the operation game adds the second, enabling a clean formal experiment of the subwar capability's effect (Paine and Tyson 2020). We select the largest accepted demand in each game. Because the games differ only in the continuation menu, the comparison isolates how the subwar capability changes the target's willingness to comply.

The model traces a common bargaining response through three outcomes: coercive ability, the prospect of peace, and the total cost of conflict. First, the subwar capability can reduce coercive ability and the coercer's payoff. Coercive ability equals punishment after resistance minus harm that compliance fails to prevent. The operation changes both components by replacing war or settlement after each target response. For weak coercers, it can add punishment where war is not credible and support a larger concession. For stronger coercers, it can replace a more damaging war after resistance while remaining profitable after compliance. The inverse trade-off can then disappear: weaker punishment induces a smaller concession, and the smaller concession exposes compliance to more harm. Threat and assurance weaken together. Coercive ability falls when the assurance loss exceeds any threat gain. The same response can make the capability privately costly. At a fixed concession, the larger menu cannot hurt the coercer. Once the target concedes less, however, the lost concession can exceed the operation's full gain and leave every coercer type weakly worse off. Modern technologies that lower the military cost of using force can therefore raise the political cost of settling disputes.

Second, the capability can increase war even though the operation itself substitutes for war. At a fixed concession, some low-cost coercers replace war with the operation, so war falls. But the equilibrium concession is not fixed. A target that expects greater harm after compliance concedes less; so does a target that expects less punishment after resistance. The smaller concession makes war more attractive after compliance and causes types that would have settled in the benchmark to fight. These types never use the operation. When the operation remains active after compliance, its expected use directly weakens assurance and lowers the concession. Once the operation is dominated after compliance, it can still replace war after resistance and weaken the threat. The target again concedes less, benchmark settlers choose war, and the added war further weakens assurance. The direct menu effect therefore reduces war, but the bargaining response can outweigh it and raise war even when the operation is never used on the equilibrium path.

Third, making the operation cheaper can increase the total resources conflict destroys. After resistance, a cheaper operation replaces war for more coercer types, reducing punishment and weakening the threat. After compliance, it can replace settlement for more types and weaken assurance. In the region studied below, both changes reduce the concession the target accepts. The smaller concession then makes war more attractive after compliance and weakens assurance further. In resource terms, replacing war with the operation saves costs, while replacing settlement with it adds costs. When the second effect dominates, expected conflict cost rises even at a fixed concession; the bargaining response raises it further.

First, the paper contributes to emerging research on subwar capability. Studies of faits accomplis explain why states seize limited gains rather than seek concessions or launch war (Tarar 2016; Altman 2017; 2020; Maass 2021). Work on hassling and gray-zone conflict shows how low-level force can avert preventive war, exploit deterrence constraints, or provoke escalation (Schram 2021; Gannon et al. 2024; Kenkel and Schram 2025). Models of flexible response explain substitution toward limited retaliation or cheaper, weaker instruments (Powell 1989; Spaniel and İdrisoğlu 2024). These studies principally examine the use, intensity, or escalation effects of limited force. We isolate the bargaining effect of the capability's availability from the direct effect of its use. The same operation enters an otherwise unchanged continuation menu after either target response, so it changes both punishment after resistance and protection after compliance. The target adjusts its concession in response. This design shows how the mere availability of subwar force can change coercive ability, war, the coercer's payoff, and the total cost of conflict, including for coercer types that never conduct the operation.

Second, the paper contributes to research on military power and coercion. The capacity to impose costs short of victory need not track the capacity to win (Slantchev 2003; 2011). Greater power also need not translate into coercive success: powerful coercers may fail when they cannot credibly promise restraint (Sechser 2010; Art and Greenhill 2018; Cebul, Dafoe, and Monteiro 2021). Related formal models show that low-level capabilities can hurt their holders through predictability, emboldening, or escalation (Schram 2022; Gannon et al. 2024; Kenkel and Schram 2025). Those models give the limited instrument to a responder and ask how it changes the challenge the responder faces. We give the subwar capability to the coercer and allow the target's concession to respond to its anticipated use. Separating war-fighting power from subwar capability then reveals why a larger military menu need not increase coercive ability. At a fixed concession, the extra option cannot hurt the coercer; in equilibrium, however, the target may concede less. The capability's political value therefore depends on whether it replaces war or settlement after resistance and compliance.

Third, the paper provides a rare formal analysis of coercive ability as a composite of threat and assurance. Coercion is widely understood to require both a credible threat against resistance and assurance that compliance will end the harm (Schelling 1966; Cebul, Dafoe, and Monteiro 2021; Pauly 2024; 2025). Pape (1996) classifies coercive strategies by how they make resistance costly or futile, capturing the threat side of this problem.

Kydd and McManus (2017) provide the closest formal analysis: a threat raises the coercer's bargaining bottom line, while a public assurance lowers it to restore agreement. Threat and assurance therefore work in opposite directions. Assurance here is an equilibrium outcome instead. The target infers it from the continuation action the coercer will find optimal once the transfer is in hand. Expected harm after resistance and after compliance therefore enter the same acceptance constraint. Because the concession also changes the post-compliance action, a weaker threat can induce a smaller concession that further weakens assurance. The model thus identifies conditions under which threat and assurance deteriorate together and traces the common cause to the target's endogenous concession.

The Framework

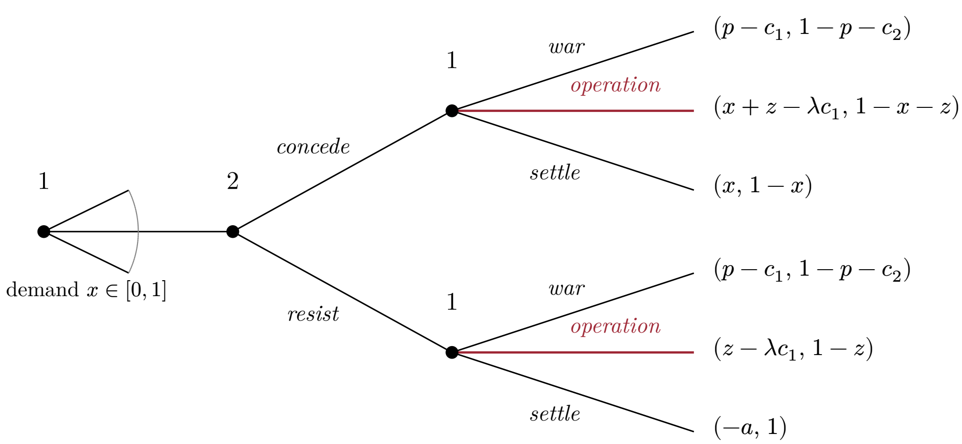

Figure 1: Coercion Game

Note: Player 1 is the coercer, player 2 the target; the coercer's payoff appears first. The colored operation branches are added by the subwar capability and absent from the benchmark. War reopens the dispute, so its payoffs do not depend on the transfer. The operation preserves the transfer, produces a bounded gain z, and costs λc_1. Settling after resistance costs the coercer a.

The analysis compares two crisis-bargaining games. They share the same players, information, demand, and target response. They differ only in the actions available to the coercer after the target responds. This design identifies how adding the operation to the coercer's menu changes punishment after resistance, protection after compliance, the demand the target accepts, and the equilibrium prospect of peace.

Figure 1 shows the common timing. The coercer first makes a take-it-or-leave-it demand. The target then concedes or resists. Finally, the coercer chooses a continuation action. The coercer's cost of war is private information, and abandoning a resisted demand carries an audience cost (Banks 1990; Fearon 1994; Schultz 1999). The benchmark continuation menu contains settlement and war. The second game adds a subwar operation after either target response.

The final stage determines both requirements of coercion. The action chosen after resistance determines the threat. The action chosen after concession determines how much protection compliance provides. A continuous demand measures the target's response to both. Unlike models in which the proposer screens an uncertain target, private information here determines the coercer's conduct after the target has moved (Fearon 1995; Powell 1996).

Players, Actions, and Timing

Two states value a disputed good at 1. The *target*, player 2, initially holds the good. The *coercer*, player 1, has a war cost c_1 drawn uniformly from $[0, \bar{c}]$. The coercer observes its cost, but the target knows only its distribution F, with density $f = \frac{1}{\bar{c}}$. A lower cost denotes a more resolute coercer. The target's war cost $c_2 \geq 0$ is common knowledge.

The coercer demands $x \in [0, 1]$. The target either concedes the demand or resists and transfers nothing. Let y be the transfer in place at the continuation stage. Thus $y = x$ after concession and $y = 0$ after resistance. The coercer chooses from the same continuation menu after either response.

Settlement leaves the transfer y in place. If the target resisted, settlement also imposes an audience cost $a > 0$ on the coercer for abandoning its demand. War gives payoffs $(p - c_1, 1 - p - c_2)$, where $p \in (0, 1)$ is the coercer's probability of victory. Because war reopens the whole dispute, these payoffs do not depend on any earlier transfer.

The second game adds a *subwar operation*. The baseline puts different operations on a common payoff scale. Let

$$S(y) = \min\{z, 1 - y\}$$

denote the resulting gain to the coercer and harm to the target. The operation costs the coercer λc_1, where $\lambda \in (0, 1)$, and produces payoffs $(y + S(y) - \lambda c_1, 1 - y - S(y))$. This reduced form places different operations on a common payoff scale. It preserves the earlier transfer and bounds the operation's effect by z.

The analysis maintains the following parameter region.

Assumption 1 (Parameter Restrictions): (i) The operation has limited reach,

$$p + c_2 + 2z \leq 1$$

and (ii) uncertainty about the coercer's war cost is sufficiently large,

$$\frac{2p + c_2}{1 - \lambda} < \overline{c}$$

A war costs the target $p + c_2$ relative to keeping the disputed good. Assumption 1(i) ensures that the operation has its full effect at every accepted demand, so $S(y) = z$ there.[1] Assumption 1(ii) implies $\overline{c} > 2p + c_2$ in the benchmark and directly imposes the stronger bound needed in the operation game. These inequalities make the target's payoff from concession decrease with the demand. Additional conditions governing interior cutoffs and the coercer's continuation menu are stated with the results that use them.

[1]If war occurs with positive probability after compliance, the post-compliance war cutoff is positive, so $x^* < p - z$ when the operation is active and $x^* < p$ when it is dominated; either way $x^* + z < h + z \leq 1 - z$. If war does not occur after compliance, then $\Omega^* = x^*$, and (6) gives $x^* = \Theta^* - \Delta^* \leq \Theta^* \leq \max\{h, z\}$, so $x^* + z \leq \max\{h + z, 2z\} \leq 1 - \min\{z, h\}$. The benchmark is the case $z = 0$.

Comments on the Model

Following Paine and Tyson (2020), we adopt the experimental perspective on model comparison. Every primitive and move is fixed except the coercer's continuation menu, so differences between the games arise from the subwar capability and the target's response to it. The open post-compliance continuation choice is necessary for the conflict consequences studied here: with perfect assurance, the operation could still soften the threat and reduce the concession, but the smaller concession could not induce further force. The comparison is nonetheless between a selected equilibrium of one game and a selected equilibrium of another, and the section on equilibrium selection states what that selection delivers and what it does not.

Reduced form. The model uses a common reduced form for different subwar operations rather than distinguishing their physical forms. The baseline sets the coercer's gain from the operation equal to the harm it imposes on the target. Corollary 1 separates these effects and shows how each enters the result. The analysis therefore treats z as a payoff effect, not as a description of what the operation physically does.

Payoff scope. The payoffs impose two scope conditions. First, general war supersedes an earlier transfer, as in standard bargaining models of war (Fearon 1995; Powell 1996). All main results are robust to allowing war to reclaim only part of an accepted concession; §F.3. Appropriative War gives the extension. Second, for tractability, the subwar operation preserves the transfer and does not trigger general war within the model's horizon. This condition makes post-compliance conduct independent of the size of the demand. After the target pays, settlement and the operation both preserve the transfer. The transfer cancels when the coercer compares those actions, so paying more cannot buy greater restraint.

Post-compliance military advantage. The baseline holds the probability of victory fixed across the target's responses. If compliance instead raises it from p to $p + \varepsilon$, even an arbitrarily small advantage makes every coercer type strictly prefer an accepted positive demand to a resisted one. §F.2. Post-Compliance Military Advantage solves the extension in pure strategies. When the operation is less harmful than advantaged war, every positive-compliance equilibrium is outcome-equivalent to pooling at its largest accepted demand. The advantage lowers that demand and coercive ability while raising war and total conflict cost in both games. The paper's strongest backfire comparison also survives: if the benchmark demand exceeds the operation-game demand by more than the operation's gain, the operation lowers peaceful extraction and the coercer's payoff while raising war and total conflict cost.

Escalation risk. If an operation triggered general war with positive probability, using it would place the transfer at risk. The operation–settlement cutoff then falls in the transfer, so a larger concession raises the probability of settlement after compliance, which is the response the fixed-effect specification removes. A small escalation risk therefore weakens the mechanism, and the weakening grows with the risk. §F. Extensions and Scope states the extended cutoff and signs both comparative statics. The model applies to operations that occur while an accepted arrangement remains in force and treats later general war as outside its horizon.

Irreversible concessions. The model treats a completed transfer as irreversible within its horizon. This focus is deliberate: the assurance problem is most severe when the target cannot reverse its concession after further force. The assumption fits vacated territory, a destroyed capability, or a surrendered position. It fits staged or reversible concessions less well, including inspection access, an enrichment cap, or a force posture. Modeling staged implementation would require another target response after the operation and lies outside the baseline.

Subwar capability. Possession is exogenous because the paper asks how having the subwar capability changes bargaining, not why states acquire it, a question studied elsewhere (Schram 2022). The model compares a coercer that has the subwar capability with one that does not; it does not model investment, disclosure, divestment, or selection into acquisition. Proposition 6 therefore compares payoffs conditional on possession rather than providing an acquisition rule. The analysis also distinguishes bargaining performance from the coercer's payoff. Coercive ability measures the accepted-demand component retained without general war, whereas ex ante payoff also values the continuation menu. Different conditions govern these two quantities.

Operating costs. The operation costs λc_1. Its cost rises with the coercer's war cost but less rapidly because $\lambda < 1$. This specification represents operations that use the same underlying capabilities as war at a smaller scale. Type heterogeneity in operating cost is load-bearing: with a type-independent cost, all types rank the operation and settlement identically after compliance, so α is zero or one except at a knife edge. The operation carries no audience cost in the baseline. Only the visible abandonment of a resisted demand does. A public demand can create such a cost, while private threats can reduce it. Alternative bargaining protocols can also change crisis outcomes (Kurizaki 2007; Fey and Kenkel 2021). Both features remain fixed across the games; the extension below varies the operation's political cost explicitly.

Prior signals. The model omits signals sent before the demand, including public statements and military threats (Fearon 1994; 1997; Schultz 1999; Slantchev 2005). We omit these features because they would be present in both games. Holding prior signals fixed leaves the continuation menu as the only source of differences. This choice separates the mechanism from informational accounts of inexpensive force. Post (2019) finds that low-cost air mobilizations reveal little resolve and can reduce the success of compellent threats. The on-path demand conveys no information in the pooling equilibria studied here. The operation instead changes expected harm after both resistance and compliance.

The Benchmark

Begin with the benchmark by setting $z = 0$. The coercer can then choose only settlement or war after either target response. We compare the games using the amount obtained without reopening the dispute.

Definition 1 (Coercive Ability): A coercer's *coercive ability* Ω is the expected portion of an accepted demand that remains with the coercer without general war. If the target concedes x, then

$$\Omega = x\,[1 - \Pr(\text{war})]$$

where $\Pr(\text{war})$ is the probability of war after concession. War reopens the dispute and reverses the transfer. Throughout, *coercive ability* refers only to this paper-specific measure. It excludes any additional amount taken through the subwar operation and all continuation costs. Proposition 6 separately reports the coercer's expected total payoff.

This definition applies to both games. The solution concept is pure-strategy perfect Bayesian equilibrium. We focus on pooling equilibria and select the largest accepted demand in each game. The results are robust to allowing pure separating demands.[2] We retain only equilibria that satisfy the intuitive criterion (Cho and Kreps 1987), and the target concedes when indifferent.[3] The convention makes the binding on-path demand attainable and makes zero an accepted demand when the target is indifferent. Under rejection at equality, the selected demand is the supremum of accepted demands and the reported outcomes are

[2]Lemma 5 in the Supporting Information establishes this robustness. Every type that does not fight after concession strictly prefers the largest accepted demand and therefore pools there. Only types that fight after both target responses can separate. Because those types give the target the same payoff after concession and resistance, their separation does not change the target's acceptance condition. The pooling equilibrium therefore attains the largest accepted demand and is coercer-optimal even when separating demands are allowed.

[3]D1 is a stronger requirement for off-path belief updating, and its verdict is symmetric across the two games. Proposition 9 in the Supporting Information shows that no equilibrium with pure target responses and an interior post-compliance war cutoff survives D1 in either game. The criterion therefore empties the class within which $x^\dagger$ and x^* are compared rather than selecting a different demand in one game. Resolute coercer types are good news for the target because concession moves them away from war, which inverts the usual logic of D1. If $c_2 > 0$, the elimination is independent of the tie rule; at $c_2 = 0$, it is tie-breaking-sensitive. We impose the intuitive criterion and do not claim that the pooling outcomes survive D1.

the corresponding limits. A dagger marks benchmark quantities. Unmarked symbols refer to the game with the operation. The Supporting Information contains all proofs.

At a fixed demand, the operation cannot mechanically reduce coercive ability. If it replaces war after compliance, the transfer survives more often. If it replaces settlement, the transfer remains in place. Any reduction in coercive ability must therefore arise because the subwar capability changes the demand the target accepts.

To solve the benchmark, start at the continuation stage. War becomes less attractive as c_1 rises, while the settlement payoff is constant in c_1. After transfer y, the coercer fights when

$$c_1 \leq \kappa^\dagger(y) = p - y + a\mathbb{1}[y = 0]$$

and settles otherwise. A larger transfer lowers this cutoff because it makes settlement more valuable.

Concession weakly raises every coercer type's continuation value. A type that settles receives x after concession instead of paying a after resistance. A type that fights receives $p - c_1$ after either response and is indifferent. Thus no type prefers resistance, and every settling type prefers the largest accepted demand.

We therefore consider a pooling equilibrium in which the demand conveys no information about c_1. Fighting types could make other demands because they are indifferent, but these deviations do not change the target's loss from war. Under the conditions below, they do not change the largest accepted demand, the probability of war, or coercive ability.

The target's acceptance decision determines the pooling demand. After resistance, the coercer fights when $c_1 \leq p + a$. After concession of x, the cutoff falls to $p - x$ because settlement now preserves the transfer and carries no audience cost. Types in $(p - x, p + a]$ therefore fight after resistance but settle after concession. Conceding costs the target x only when war does not reopen the dispute, while the benefit is the war loss avoided for these types. The target accepts when

$$x(1 - F(p - x)) \leq (p + c_2)(F(p + a) - F(p - x))$$

The largest accepted demand makes this inequality bind. Under the uniform distribution, the binding condition is a quadratic whose positive root is $x^\dagger$.

Proposition 1 (Benchmark Equilibrium and Coercive Ability): Suppose the relevant war cutoffs are interior at every accepted demand. Under Assumption 1, a pooling equilibrium exists in which every type demands $x^\dagger$ and the coercer fights if $c_1 \leq \kappa^\dagger(x^\dagger)$. Among equilibria with pure target responses, this is the unique outcome at the largest accepted demand, and it satisfies the intuitive criterion. The coercer fights with probability $F(p+a)$ after resistance and $F\big(\kappa^\dagger(x^\dagger)\big)$ after concession. Its coercive ability is given by (2), and settlement after compliance occurs with probability $\alpha^\dagger = 1 - F\big(\kappa^\dagger(x^\dagger)\big)$.

The largest accepted demand is

$$x^\dagger = \frac{1}{2}\left[-(\bar{c} - 2p - c_2) + \sqrt{(\bar{c} - 2p - c_2)^2 + 4(p + c_2)a}\right] \tag{1}$$

Proposition 1 determines both the demand and continuation behavior. Types below the cutoff fight after concession, while types above it settle. Greater military power raises the cutoff and makes war optimal for a larger set of coercer types.

$$\Omega^\dagger = \underbrace{(p + c_2)\, F(p + a)}_{\text{punishment after resistance}} - \underbrace{(p + c_2)\, F\big(\kappa^\dagger(x^\dagger)\big)}_{\text{harm not prevented by compliance}} \tag{2}$$

Equation (2) writes coercive ability as one difference. The first term is the expected punishment produced by resistance. The second is the expected harm that compliance fails to prevent. A larger concession lowers the post-compliance war cutoff, makes settlement more likely, and reduces the second term. Greater power raises the first term and raises coercive ability as a whole, so whatever protection compliance loses is outweighed. The benchmark thus reproduces the expectation that power strengthens the threat faster than it erodes assurance (Cebul, Dafoe, and Monteiro 2021). The next section asks whether the subwar capability behaves the same way.

Adding Subwar Capability

Now set $z > 0$ and solve the enlarged continuation menu. The payoffs from war, the operation, and settlement have slopes -1, $-\lambda$, and 0 in c_1. Because $0 < \lambda < 1$, the optimal action changes from war to the operation and then to settlement as the coercer's war cost rises. The relevant cutoffs are

$$\kappa_W(y) = \frac{p - y - S(y)}{1 - \lambda} \quad \text{and} \quad \kappa_L(y) = \frac{S(y) + a\mathbb{1}[y = 0]}{\lambda} \tag{3}$$

If $\kappa_W(y) < \kappa_L(y)$, low-cost types fight, intermediate types use the operation, and high-cost types settle. If the inequality is reversed, the operation is dominated and the coercer chooses between war and settlement.

Let $W(y)$ be the probability of war after transfer y, and let $L(y)$ be the probability of either war or the operation. When the operation is optimal for some types, $W(y) = F(\kappa_W(y))$ and $L(y) = F(\kappa_L(y))$. When the operation is dominated, define $\kappa_S(y) = p - y + a\mathbb{1}[y = 0]$ and set $W(y) = L(y) = F(\kappa_S(y))$. The same notation therefore covers both continuation menus.

Before solving for the demand, hold transfer y fixed and compare threat and assurance. This exercise does not assume that the equilibrium demands are equal. Whenever the operation is active, $\kappa_W(y) < \kappa_S(y) < \kappa_L(y)$. Types between the first two cutoffs replace war with the operation. When war is more harmful, this substitution weakens the threat after resistance and strengthens assurance after compliance. Types between the last two cutoffs replace settlement with the operation. That substitution strengthens the threat after resistance and weakens assurance after compliance. The action replaced, not the operation alone, therefore determines how the target evaluates resistance and compliance.

The cutoffs determine when the operation enters the coercer's menu. When $S(y) = z$, it is optimal for some types after resistance if and only if $z + a > \lambda(p + a)$, and after concession of x if and only if $z > \lambda(p - x)$. Greater reach or lower relative cost makes the operation more attractive after either response. A larger audience cost also favors the operation after resistance because using it avoids the cost of backing down. A larger concession favors the operation over war after compliance because the operation preserves the transfer and war does not.

Having solved the coercer's continuation choice, turn to the target's response. The target concedes x when its expected loss after resistance is at least as large as its expected loss after compliance:

$$\begin{aligned}&(p+c_2)W(0)+z(L(0)-W(0))\\&\quad\geq x(1-W(x))+(p+c_2)W(x)+S(x)(L(x)-W(x))\end{aligned} \tag{4}$$

The left side of (4) is expected punishment after resistance. On the right, the first term is the transfer that survives unless war reopens the dispute. The remaining terms are harm that compliance does not prevent. The target's acceptance decision therefore depends on continuation behavior after both responses. The equilibrium demand is the largest x that satisfies this inequality.

Let α be the probability of settlement after compliance in the pooling equilibrium. A closed-form solution requires four additional conditions. The war cutoffs and the post-compliance operation–settlement cutoff are interior. The operation is active after compliance. Every nonfighting type uses the operation after resistance, so $\kappa_L(0) \geq \bar{c}$ or $z+a \geq \lambda\bar{c}$. Finally, $\alpha > 0$.

Each condition has a separate role. Interior cutoffs make action probabilities linear under the uniform distribution. Activity after compliance ensures that all three continuation actions matter. The condition on $\kappa_L(0)$ sets $L(0)=1$, so every type that does not fight after resistance uses the operation. A positive α leaves some types that use the operation after resistance willing to settle after compliance. Compliance protects the target from the operation for exactly these types. Assumption 1(i) keeps $S(y)=z$ at accepted demands. These restrictions produce the closed form below. The substitution argument above does not require them.

Proposition 2 (Equilibrium with a Subwar Operation): Under Assumption 1 and the additional conditions above, a pooling equilibrium exists in which every type demands x^* and war occurs with probability $W(x^*)$. Among equilibria with pure target responses, this is the unique outcome at the largest accepted demand. This pooling equilibrium satisfies the intuitive criterion.

When $\alpha > 0$, the largest accepted demand is

$$x^* = \frac{1}{2}\Big[-((1-\lambda)\bar{c} - 2p - c_2 + 2z) + \sqrt{((1-\lambda)\bar{c} - 2p - c_2 + 2z)^2 + 4(1-\lambda)\bar{c}z\alpha}\Big] \tag{5}$$

The constant term inside the quadratic for x^* is proportional to $z\alpha$. The term is positive only when compliance moves a positive mass of coercer types from the operation to settlement. If $\alpha = 0$, every type uses force after compliance, so compliance provides no protection from the operation and the target accepts no positive demand. The next result determines α and explains why the accepted transfer may fail to increase it.

Consider the operation–settlement choice after compliance. In the benchmark, a larger transfer raises the settlement payoff but leaves the war payoff unchanged. It therefore makes settlement more likely. With the operation, both settlement and the operation preserve the transfer. Their payoff difference is $S(y) - \lambda c_1 - C(y)$, where $C(y)$ is any additional operating cost. A larger transfer changes this comparison only if it changes the operation's net return $S(y) - C(y)$.

Proposition 3 (Settlement After Compliance): Suppose the operation takes $S(y)$ and costs $\lambda c_1 + C(y)$ after transfer y. At any accepted demand at which the operation is active and the operation–settlement cutoff is interior, $\kappa_L(y) = \frac{S(y)-C(y)}{\lambda}$ and its derivative is $\frac{S'(y)-C'(y)}{\lambda}$. In this model, $S(y) = z$ and $C(y) = 0$

at every accepted demand, so $\kappa_L(y) = \frac{z}{\lambda}$ is constant in the demand and $\alpha = 1 - F\left(\frac{z}{\lambda}\right)$. Settlement becomes less likely as z rises and more likely as λ rises.

Proposition 3 characterizes one continuation boundary at an accepted demand, not the full acceptance decision. Under the fixed-effect specification, a larger transfer can move types from war to the operation through κ_W. It cannot move them from the operation to settlement because κ_L is constant. With a general benefit or cost technology, $S'(y) - C'(y)$ determines that second response. The target still chooses the demand according to (4). In Figure 4, the constant cutoff appears as the horizontal upper boundary of the region where the operation replaces settlement.

Spaniel (2023) studies the neighboring case in which a limit on what can be taken is itself pacifying, and finds a non-monotonic relation: war is least likely at moderate indivisibility, because an extreme one leaves no acceptable agreement (his Propositions 1 and 2). His limit is on the divisibility of the stake rather than on the instrument. Proposition 5 below is the mirror image of the same comparison: removing a binding restriction on what the coercer may take after compliance can leave every coercer type weakly worse off.

In the baseline, z is at once the coercer's gain and the target's harm. To separate their effects, let an operation impose $d > 0$ on the target and deliver $g \geq 0$ to the coercer, with the two unlinked. Suppose the relevant post-resistance cutoffs are interior.

Corollary 1 (Gain and Harm): The coercer's gain alone determines settlement after compliance: $\alpha = 1 - F\left(\frac{g}{\lambda}\right)$, independent of the target's harm. If $g = 0$ and the operation is active after resistance, the subwar capability raises the accepted demand and coercive ability if and only if $d > \lambda h$.

The target's loss d governs how much harm each operation does. The coercer's gain g governs whether compliance protects at all. An operation that provides no benefit still costs

λc_1 to conduct, so no type conducts it once the transfer is in hand. The assurance channel therefore depends on the operation's benefit to the coercer, not on target harm alone.

Now return to resistance and ask whether the operation replaces war. If the operation is active after resistance, war occurs with probability $F\left(\frac{p-z}{1-\lambda}\right)$; otherwise it occurs with probability $F(p+a)$. The operation is active, and lowers the probability of war, if and only if $z+a > \lambda(p+a)$. It never raises war after resistance.

The baseline assigns no audience cost to the operation, so it provides a way to avoid backing down after resistance. Suppose instead that using the operation imposes an audience cost $a_L \in [0, a]$ after either response. Two boundaries then govern the extension. The operation replaces war after resistance if and only if $z + a - a_L > \lambda(p+a)$, and it remains in the post-compliance menu if and only if $a_L < z - \lambda(p-x)$; past that boundary compliance fully protects against the operation, although war may still follow. §F.4. The Operation's Audience Cost states the cutoffs and the modified acceptance condition. The two political costs have distinct roles: a makes the operation relatively attractive after resistance, while a_L makes it less attractive after both responses. The comparison is therefore conditional on their joint configuration, not on publicity alone.

The probabilities π and α describe only two continuation actions. The resistance-stage comparison holds the history fixed and does not determine the equilibrium probability of war after the target complies, because the subwar capability can change the accepted demand. To evaluate the target's full decision, convert every continuation outcome into expected loss. Let Θ^* denote punishment after resistance and Δ^* denote harm not prevented by compliance.

Remark 1 (Coercive Ability as Threat Minus Assurance Failure): If a common accepted demand leaves the target indifferent, coercive ability equals expected punishment after resistance, Θ^*, minus harm not prevented by compliance, Δ^*. It is zero exactly when $\Theta^* = \Delta^*$. This identity follows from (4) and does not require the closed-form equilibrium.

$$\begin{aligned}\Omega^* = &\underbrace{(p + c_2)W(0) + z(L(0) - W(0))}_{\text{threat: punishment after resistance } =:\Theta^*} \\ &- \underbrace{(p + c_2)W(x^*) + S(x^*)(L(x^*) - W(x^*))}_{\text{assurance failure: harm not prevented by compliance } =:\Delta^*}\end{aligned} \tag{6}$$

At the largest common accepted demand, the target is indifferent. Rearranging (4) gives $x^*(1 - W(x^*)) = \Theta^* - \Delta^*$. Let $\Theta^\dagger$ and $\Delta^\dagger$ be the corresponding benchmark quantities from Proposition 1 and (2). The difference between the games is

$$\Omega^* - \Omega^\dagger = \left(\Theta^* - \Theta^\dagger\right) - \left(\Delta^* - \Delta^\dagger\right) \tag{7}$$

The operation increases coercive ability when its effect on punishment exceeds its effect on post-compliance harm. It reduces coercive ability when the reverse inequality holds. The operation weakens the threat when it lowers Θ^* and weakens assurance when it raises Δ^*. Which effect occurs depends on whether the operation replaces war or settlement after each target response. Setting $z = 0$ removes the operation terms and recovers (2). The following sections compare these two changes.

When Threat and Assurance Weaken Together

The operation's effect depends on the action it replaces after each target response. Replacing settlement after resistance increases punishment and strengthens the threat. Replacing a more damaging war after resistance reduces punishment and weakens the threat. After compliance, replacing settlement adds harm and weakens assurance, while replacing a more damaging war removes harm and strengthens assurance. Different coercer types can make different substitutions, so the four effects may reinforce or offset one another.

The shaded regimes include the target's endogenous demand response. To see the underlying continuation choices, first hold the transfer fixed. Adding the operation can change behavior in only two ways. Some coercer types use it instead of war, reducing the target's loss from h to z. Others use it instead of settlement, increasing the target's loss from zero to z. The operation's net effect at any history is the harm added by former settlers minus the harm removed by former fighters.

With interior cutoffs and a uniform cost distribution, these two groups have a simple relation: for every type shifted out of war, $\frac{1-\lambda}{\lambda}$ types are shifted out of settlement. Expected harm therefore rises when $z > \lambda h$, because the harm added by former settlers dominates, and falls when $z < \lambda h$, because replacing war dominates. The equality $z = \lambda h$ marks the balance between the two groups. §D.10. Interior Switch Accounting gives the cutoff algebra.

The same logic applies after resistance and compliance, but the number of types switching at each history differs. The audience cost makes the operation more attractive after resistance because it avoids the penalty for backing down. The accepted transfer makes it more attractive after compliance because the coercer keeps what the target has already paid. When the operation adds harm, switching after resistance strengthens coercion and switching after compliance weakens it. When the operation replaces a more damaging war, these effects reverse. Coercive ability depends on which switch is larger.

Two equilibrium responses qualify this simple comparison. One switching group may be exhausted. In the closed-form region, every type that avoids war after resistance already uses the operation, whereas some types still settle after compliance. The target can also respond by accepting a smaller demand, which may produce more war after compliance. Together, these responses allow the operation to weaken both threat and assurance. §D.10. Interior Switch Accounting gives necessary and sufficient conditions for this joint loss within the closed-form region.

The link between the two components runs in one direction. Punishment after resistance is evaluated at $y = 0$, so Θ^* does not vary with the demand, and assurance failure cannot weaken the threat. The reverse channel is active. A smaller Θ^* lowers the largest accepted demand, a lower demand raises $\kappa_W(x^*)$, and the additional war after compliance raises Δ^*. A weaker threat is therefore converted into a larger assurance failure rather than offset by one. The conversion compounds, because the war it induces lowers the demand again. In the closed-form region, differentiating the binding acceptance condition gives

$$\frac{\partial x^*}{\partial \Theta^*} = \frac{1}{1 - W(x^*) - \frac{h-z-x^*}{(1-\lambda)\bar{c}}}$$

which exceeds one whenever $x^* < h - z$. The concession falls by more than the threat loss alone would produce.

Equation (7) remains the general comparison. Coercive ability rises when punishment increases more than post-compliance harm and falls when the reverse is true. The accepted demand, the probability of war, and coercive ability consequently need not move together.

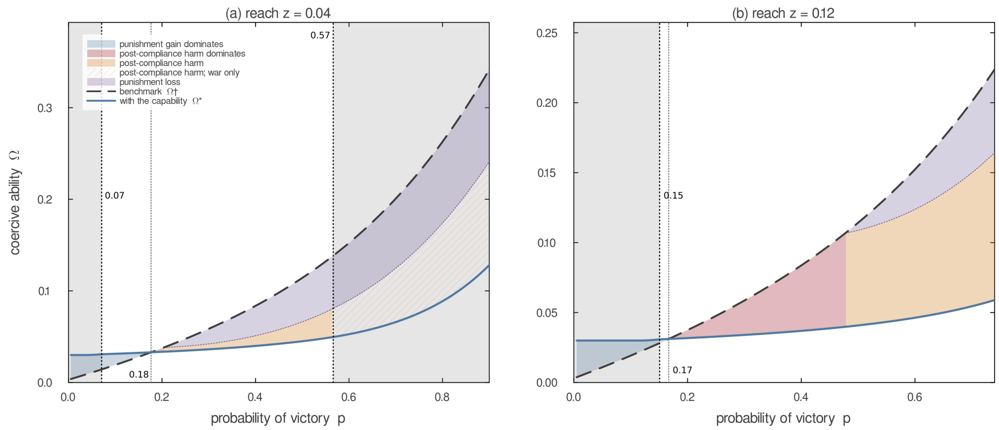

Figure 2: Threat and Assurance Can Weaken Together

Note: Coercive ability in the benchmark (dashed) and with the subwar capability (solid) as the coercer's probability of victory varies. Blue marks where stronger punishment dominates added post-compliance harm; red marks where post-compliance harm dominates despite stronger punishment. Where both channels reduce coercive ability, orange and purple decompose the loss into post-compliance harm and weaker punishment. Hatching marks post-compliance harm produced entirely by war induced by the smaller demand after the operation exits the path. Grey lies outside the closed-form region. Parameters: $c_2 = 0.02$, $a = 0.30$, $\lambda = 0.08$, c_1 uniform on $[0, 2]$; $z = 0.04$ in panel (a) and $z = 0.12$ in panel (b).

Figure 2 illustrates this comparison at two values of the operation's reach. In both panels, greater punishment after resistance initially outweighs added post-compliance harm. As power rises, the coercive-ability curves cross before the punishment comparison does: post-compliance harm first becomes the dominant loss, and only later do both channels reduce coercive ability. The analytical conditions, not the plotted slices, carry the result.

The two panels also differ in what produces the loss. At the smaller reach, no coercer type eventually chooses the operation after compliance; beyond that point the loss comes from weaker punishment after resistance together with the war that the smaller demand no longer prevents. At the larger reach, post-compliance harm combines the operation's direct effect with additional war from the smaller demand. Proposition 4 gives the exact comparison. The panels illustrate its channels but do not establish their prevalence.

The fixed-effect specification makes the assurance response especially clear. After compliance, a larger transfer raises the payoffs from settlement and the operation by the same amount. It leaves the operation–settlement cutoff unchanged, although it can still move types from war to the operation. Proposition 3 shows that a general benefit or cost technology restores a response through $S'(y) - C'(y)$.

Both equilibrium outcomes admit closed forms, allowing an exact comparison of coercive ability.

Proposition 4 (Coercive Ability in Closed Form): If the benchmark war cutoffs are interior at every accepted demand, $\Omega^\dagger = \frac{h(x^\dagger + a)}{\bar{c}}$. Under the conditions for (5), $\Omega^* = \frac{x^*(h-z)}{(1-\lambda)\bar{c}} + z\alpha$. The subwar capability lowers coercive ability if and only if:

$$\frac{h\big(x^\dagger + a\big)}{\bar{c}} > \frac{x^*(h-z)}{(1-\lambda)\bar{c}} + z\alpha \tag{8}$$

Because (1) and (5) express both demands in primitives, (8) is an exact condition. In the benchmark, coercive ability equals the target's loss from war, h, multiplied by the sum of the demand and audience cost and divided by the support of the coercer's war cost. The audience cost enters because it makes abandoning a resisted demand less attractive.

With the operation, the two terms in Ω^* correspond to two groups whose behavior differs across the target's responses. The probability of war falls by $\frac{x^*}{(1-\lambda)\bar{c}}$ between resistance and compliance. For the types responsible for this difference, compliance replaces war, which costs the target h, with the operation, which costs z. Their contribution to coercive ability is therefore $\frac{x^*(h-z)}{(1-\lambda)\bar{c}}$. A second group uses the operation after resistance but settles after compliance. This group has mass α, and compliance protects the target from harm z, producing the term $z\alpha$.

The second term is single-peaked in reach because $z\alpha = z\big(1 - \frac{z}{\lambda\bar{c}}\big)$, which is maximized at $z = \frac{\lambda\bar{c}}{2}$. Greater reach increases the harm avoided for every type in the second group, but it also shrinks that group by making the operation preferable to settlement for more types after compliance. The two terms may move in opposite directions, so neither reach nor the probability of war alone signs the comparison. The resistance-stage condition determines when war after resistance falls, but the target also accounts for operations that replace settlement.

Remark 2 (Power): In the closed-form region, the protection term $z\alpha$ depends on reach, relative cost, and the spread of war costs but not on the probability of victory. Total Ω^* still depends on power through both x^* and $h - z$, so the derivative comparison with $\Omega^\dagger$ is not signed in general.

The same cutoff identifies complete coercive failure.

Remark 3 (Complete Coercive Failure): Coercive ability is zero if and only if $z \geq \lambda\bar{c}$. When the inequality holds, the operation–settlement cutoff after compliance is at or above the top of the type distribution, so even the highest-cost coercer type prefers the operation to settlement after compliance. Compliance then provides no protection from the operation, $\alpha = 0$, the target refuses every positive demand, and $x^* = \Omega^* = 0$.

The expression $\alpha = 1 - \frac{z}{\lambda\bar{c}}$ applies only for $z \leq \lambda\bar{c}$; above that value the cutoff leaves the support of the war-cost distribution and $\alpha = 0$. Remark 3 itself does not depend on the closed-form quadratic. Its proof uses only the target's gain from conceding an arbitrarily small demand and the monotonicity of the acceptance margin, both established in the Supporting Information: when $\alpha = 0$ that gain is zero, and the margin is strictly decreasing at every larger demand.

The same equilibrium response can make the subwar capability privately costly. The relevant comparison is whether the operation's direct gain compensates for the smaller accepted demand.

Proposition 5 (The Settlement Forgone Exceeds the Operation's Gain): Suppose the benchmark post-compliance war cutoff is interior and the operation-game closed-form conditions hold. The benchmark demand exceeds the operation-game demand plus the operation's full direct gain, $x^\dagger > x^* + z$, if and only if (9) holds. Under the same condition, every coercer type is weakly worse off with the operation, a positive measure is strictly worse off, and war after compliance is strictly more likely with the operation than in the benchmark.

$$\big(x^\dagger - z\big)\big(1 - W\big(x^\dagger - z\big)\big) > \frac{\big(x^\dagger - z\big)(h - z)}{(1-\lambda)\bar{c}} + z\alpha \tag{9}$$

Condition (9) applies the target's acceptance rule at $x^\dagger - z$. At that demand, adding the operation's full gain would restore the benchmark settlement. The condition says that the transfer surrendered exceeds the harm avoided, so the target refuses that demand and every larger one.

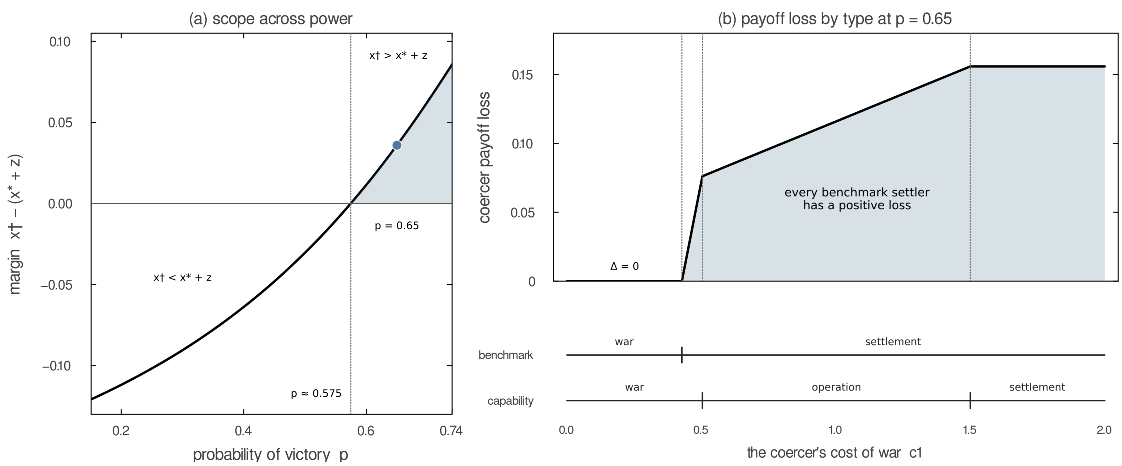

Figure 3: When the Forgone Settlement Exceeds the Operation's Gain

Note: Panel (a) plots $m(p) = x^\dagger(p) - x^*(p) - z$ over the interval where Proposition 5′s other premises hold; the condition begins at $p \approx 0.57$. Panel (b) fixes $p = 0.65$ and plots the benchmark payoff minus the operation-game payoff by coercer type; the aligned intervals show each game's post-compliance action. Parameters: $c_2 = 0.02$, $a = 0.30$, $\lambda = 0.08$, $z = 0.12$, and c_1 uniform on $[0, 2]$.

Figure 3 shows the condition and who bears the loss. Types that fight in both games are unaffected. Every benchmark settler loses, whether the smaller concession leads it to fight, conduct the operation, or settle for less. The condition also implies $\kappa_W(x^*) > \kappa^\dagger(x^\dagger)$, so war after compliance is more likely in the same region.

The next result separates the value of the larger continuation menu from the cost imposed by the target's response.

> Proposition 6 (The Coercer's Expected Payoff): If the relevant cutoffs are interior and the operation is active after concession, the coercer's ex ante payoffs are given by (10). Each payoff equals the accepted demand plus the value of the continuation menu. The coercer is worse off with the operation if and only if $V^\dagger > V^*$.

$$\begin{aligned} V^\dagger &= x^\dagger + \frac{(p - x^\dagger)^2}{2\overline{c}} \\ V^* &= x^* + \frac{(p - x^* - z)^2}{2(1-\lambda)\overline{c}} + \frac{z^2}{2\lambda\overline{c}} \end{aligned} \tag{10}$$

Let $Q_0(y) = y + \frac{(p-y)^2}{2\overline{c}}$ denote the benchmark continuation value at transfer y. The payoff difference can be written as

$$V^* - V^\dagger = \underbrace{\frac{[\lambda(p - x^*) - z]^2}{2\lambda(1-\lambda)\overline{c}}}_{\text{menu at the same transfer}} + \underbrace{Q_0(x^*) - Q_0(x^\dagger)}_{\text{endogenous-demand response}}$$

The menu term is nonnegative because every benchmark action remains available. The demand-response term is negative when $x^* < x^\dagger$. The subwar capability hurts the coercer exactly when the second effect dominates.

Subwar Capability and the Prospect of Peace

The preceding results determine how much the target concedes. They also change the continuation outcome. Figure 4 compares post-compliance actions across the two games for each combination of power and war cost. Among low-cost types, the operation can replace war. Among intermediate types below κ_L, it replaces settlement. A third change arises from the target's response. When the subwar capability lowers the accepted demand, types near the benchmark war cutoff receive too little to settle and instead choose war. These types never use the operation. The subwar capability changes their behavior only because it reduces the concession.

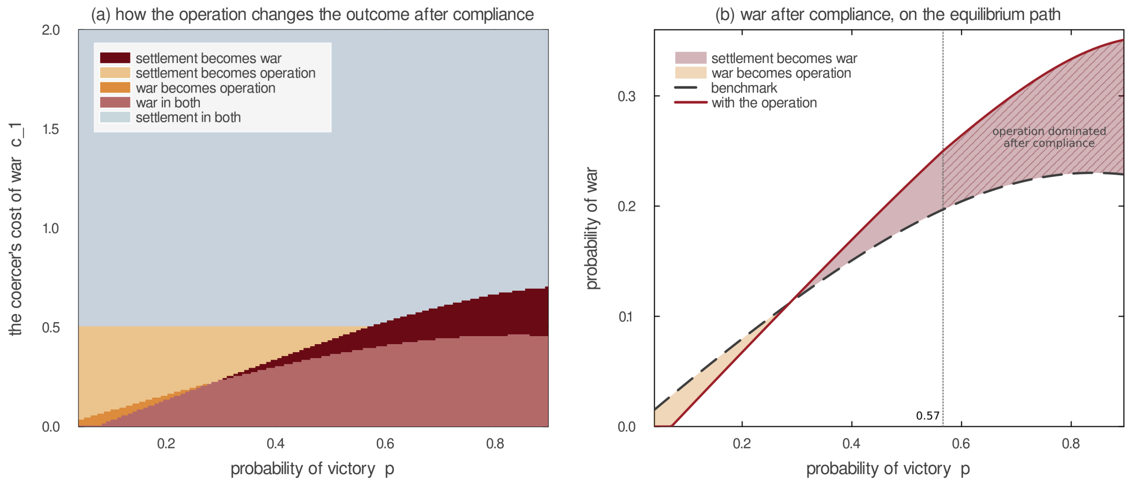

Figure 4: Subwar Capability Can Replace War Yet Increase It

Note: Panel (a) colors each combination of the coercer's probability of victory and war cost by the change in the post-compliance outcome when the operation becomes available, at the calibration of Figure 2 with $z = 0.04$. The operation region's upper boundary is horizontal because $\kappa_L = \frac{z}{\lambda}$ contains neither p nor the demand. Panel (b) plots the probability of war after compliance on each game's equilibrium path: below $p \approx 0.29$ the operation replaces war; above it, the smaller accepted demand pushes types that would have kept the benchmark settlement into war. Hatching marks the range above $p \approx 0.57$, where the operation is dominated after compliance. There the post-compliance menu is the benchmark menu evaluated at a smaller demand, and the entire increase follows from the weaker punishment the target expects after resistance.

The direct menu effect and the target's response move war in opposite directions. At a fixed concession x, an active operation lowers the war boundary from $p - x$ to $\kappa_W(x)$ and replaces war for some low-cost types. Equilibrium adds a reinforcing response through the concession. When the operation weakens punishment after resistance, holding out becomes less costly and the target concedes less. The smaller concession makes war more attractive after compliance. The resulting increase in post-compliance war weakens assurance, which further reduces what the target will concede. The equilibrium demand x^* incorporates this feedback. Algebraically, the relevant cutoffs are $\kappa^{\dagger}(x^{\dagger}) = p - x^{\dagger}$ and $\kappa_W(x^*) = \frac{p-x^*-z}{1-\lambda}$. War rises when the concession response moves the second cutoff above the first, causing benchmark settlers to fight.

Panel (b) plots the benchmark as the dashed black curve and the game with the operation as the solid red curve. To the left of their crossing at $p \approx 0.29$, the red curve lies below the benchmark: the pale gap shows the reduction in war as the operation replaces it. To the right of the crossing, the red curve lies above the benchmark: the shaded gap shows the additional war caused by the smaller concession. Between the crossing and the dotted line at $p \approx 0.57$, some coercer types still use the operation after compliance. In the hatched region to the right of the dotted line, the operation is dominated after compliance, so war rises even though the operation is never used on the equilibrium path. Additional war therefore means that more types choose war than in the benchmark, not that an operation occurs before war.

The unhatched and hatched ranges show two versions of this feedback. Between the crossing and $p \approx 0.57$, the operation remains available after compliance. Anticipated operation weakens assurance directly, so the target concedes less. The smaller concession then sends marginal types to war. To the right of $p \approx 0.57$, no type uses the operation after compliance. Instead, the operation replaces war after resistance and weakens the threat. The target responds by conceding less. That smaller concession sends benchmark settlers to war, so compliance prevents less harm and assurance weakens. This assurance loss reinforces the target's unwillingness to concede. These effects are jointly determined in equilibrium, not a temporal sequence. In both ranges, the types drawn into war never use the operation.

Subwar Capability and the Cost of Conflict

The subwar capability changes the total resources destroyed by conflict, not only the probability of war. Let K denote expected total conflict cost. Settlement costs nothing, war costs the two states $c_1 + c_2$, and the operation costs λc_1. The demand and the operation's gain transfer value between the states, so joint welfare is $1 - K$.

Let $d = x^\dagger - x^*$ denote the concession lost when the subwar capability becomes available. Holding the operation's primitives fixed, let $\Delta K(d)$ compare benchmark total conflict cost with total conflict cost under the operation after a concession $x^\dagger - d$. Thus $\Delta K\big(x^\dagger - x^*\big) = K^* - K^\dagger$, while $\Delta K(0)$ is the direct cost effect if the concession did not change. A positive ΔK means that the subwar capability makes conflict more costly. Suppose $d \geq 0$, the relevant post-compliance cutoffs are interior, and the operation reaches some benchmark settlers, $\frac{z}{\lambda} > \kappa^\dagger\big(x^\dagger\big)$. Let $d_W = z - \lambda\kappa^\dagger\big(x^\dagger\big)$ and suppose $0 < d_W < x^\dagger$.

On an active interior branch, let $r(x) = p - x - z$. The direct effect of lowering the operation's relative cost is governed by

$$\left(\frac{z}{\lambda}\right)^2 > \frac{r(x)(r(x) + 2c_2)}{(1-\lambda)^2}. \tag{11}$$

Proposition 7 (Cheaper Operations Can Raise Total Conflict Cost): Under the preceding conditions, $\Delta K(d)$ is strictly increasing in the lost concession. If $\Delta K(0) < 0$, a unique $d_K \in (0, d_W)$ separates lower from higher total conflict cost; if $\Delta K(0) \geq 0$, every $d > 0$ raises total cost. Thus total cost can rise while war falls, and any $d \geq z$ raises total cost. On an active interior branch, lowering λ raises total conflict cost at a fixed concession if and only if (11) holds. Operation activity implies this condition when $c_2 = 0$. If it holds at $x = x^*$ and $\frac{\partial x^*}{\partial \lambda} \geq 0$, then $\frac{\partial K^*}{\partial \lambda} < 0$.

At an unchanged concession, the subwar capability can lower total conflict cost by replacing war, but it raises cost when it replaces settlement. This direct substitution is the only possible source of cost savings. Once the target concedes less, total conflict cost rises monotonically. While the operation remains active, a smaller concession moves coercer types from the operation into war. If the operation becomes dominated, a further reduction moves additional types from settlement into war. The bargaining response therefore always makes conflict more costly.

If the direct substitution initially saves resources, those savings survive only while the concession loss remains below d_K. This cost threshold lies below the war threshold because, when the two games produce the same probability of war, the operation still replaces settlement for some coercer types and adds λc_1 in costs. War frequency can therefore improve after the subwar capability has already increased the total cost of conflict.

Making the operation cheaper changes both sides of the target's acceptance decision. After resistance, lowering λ expands the operation at the expense of war. When war is more harmful, resistance becomes less costly and the threat weakens. After compliance, two changes oppose each other. Replacing war with the operation strengthens assurance, but replacing settlement with the operation weakens it. The settlement substitution dominates throughout Figure 5's range, so assurance weakens on net. The weaker threat and weaker assurance both reduce the concession. That response makes war more attractive after compliance and further weakens assurance.

The same substitutions determine total conflict cost. Replacing war with the operation saves resources, while replacing settlement with it adds costs where the benchmark would have produced none. Condition (11) states when the second effect dominates at a fixed concession. If $c_2 = 0$, operation activity itself makes the settlement-side increase larger than the war-side saving. The smaller equilibrium concession raises cost further by moving additional types from the operation into war. As λ falls from 0.21 to 0.061 in Figure 5, cost at the benchmark concession rises from 0.051 to 0.087. Allowing the concession to adjust raises equilibrium cost from 0.057 to 0.138, while benchmark cost remains 0.050.

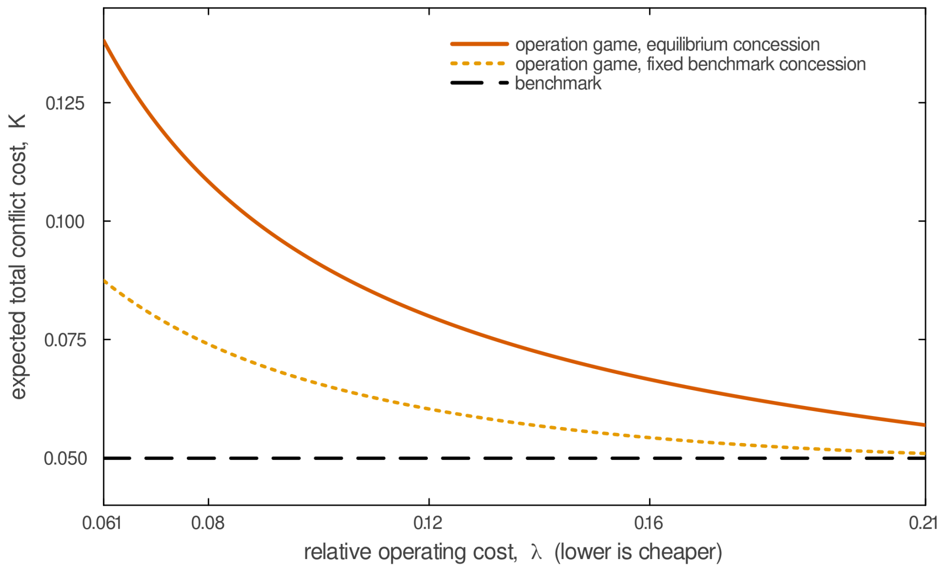

Figure 5: Cheaper Operations Can Raise Total Conflict Cost

Note: The solid red curve plots equilibrium expected total conflict cost with the operation. The dotted orange curve holds the concession at its benchmark level, isolating the direct effect of changing λ. The dashed black line is benchmark cost. Lower λ means a cheaper operation. Over the displayed interval, lowering λ weakens the threat, weakens assurance on net, and reduces the accepted concession. The interval satisfies the closed-form premises, (11), and $\frac{\partial x^*}{\partial \lambda} > 0$. Parameters: $p = 0.65$, $c_2 = 0.02$, $a = 0.30$, $z = 0.12$, and c_1 uniform on $[0, 2]$.

Scope of the Equilibrium Comparison

Part of the analysis does not depend on equilibrium selection. If the operation takes a fixed amount and is active after an accepted demand, the coercer settles if and only if $c_1 > \frac{z}{\lambda}$. That cutoff does not depend on the demand or off-path beliefs, and Proposition 8 proves it for every equilibrium. The threat-minus-assurance identity in Remark 1 is free of the closed form in the same way: it requires only a common accepted demand that leaves the target indifferent. Both claims are conditional in one respect. They apply once a demand at which the operation is active has been conceded. Different equilibria can place different demands on the path, change the types associated with each demand, or include accepted demands at which the operation is dominated, so the unconditional settlement rate and coercive ability are not fixed by the cutoff alone.

Both pooling equilibria satisfy the intuitive criterion. That restriction removes a type from an unexpected demand only when no sequentially rational target response lets the type match its equilibrium payoff. In the baseline, allowing pure separating demands does not alter the selected outcome: Lemma 5 in the Supporting Information shows that only types that fight after both target responses can separate, and their demands do not affect the target's acceptance condition. This outcome robustness rests on an indifference, because war pays the coercer the same amount after resistance and compliance. A post-compliance military advantage removes that indifference. Every type strictly prefers an accepted positive demand to a resisted one. When the operation is less harmful than advantaged war, types in any positive-mass lower pool would have to fight after compliance, which would make the target resist that demand. §F.2. The extension therefore shows that every positive-compliance pure-strategy equilibrium is outcome-equivalent to pooling. Under the stated demand condition, the coercive-ability, coercer-payoff, war, and total-cost comparisons all survive.

What the comparison selects in each game is the coercer-optimal equilibrium within the class analyzed. The Supporting Information establishes that every type's equilibrium payoff is its optimized continuation payoff at the largest accepted demand, which increases in that demand. The selection is not innocuous. A worst-case selection would make the comparison vacuous, because each game also has an equilibrium satisfying the intuitive criterion in which the target refuses every positive demand and $\Omega^\dagger = \Omega^* = 0$.[4] The comparison is deliberately best case against best case.

[4]Fix a belief concentrated on coercer types that take the same continuation action after either target response. Conceding x then leaves the target exactly x worse off than resisting, so refusal is strictly optimal at every positive demand, and no type is equilibrium dominated by the deviation. These are the types that settle after either response in the benchmark, which exist whenever $p + a < \bar{c}$, and the types that use the operation after either response in the operation game, which exist whenever $z > \lambda p$.

Conclusion

Existing accounts cast threat and assurance as an inverse trade-off. This model shows that they can weaken together. When a subwar operation reduces punishment after resistance, the target concedes less. The smaller concession makes general war more attractive after compliance, so compliance prevents less harm and assurance also weakens. This response can hurt strong coercers that already obtain favorable settlements, induce war among types that the benchmark settlement would have pacified, and leave every coercer type weakly worse off when the operation cannot compensate for the lost concession. Lower operating costs can activate the same chain and increase total conflict cost further.

These results also clarify why a subwar capability's value in deterrence need not carry over to compellence. Both require punishment to be conditional, but the compellence problem here asks the target to make a concession and then rely on the coercer not to use the operation. Because the operation remains available after compliance, it can strengthen punishment after resistance while weakening the protection that compliance buys. A low-cost option may therefore support a deterrent threat yet weaken a compellent bargain. The effect of a subwar capability on compellence cannot be inferred from threat credibility alone (Schelling 1966; Kydd and McManus 2017; Cebul, Dafoe, and Monteiro 2021).

The analysis also qualifies the usual option-value logic. At a fixed transfer, adding the operation cannot hurt the coercer because every benchmark action remains available. Once the target chooses the transfer in anticipation, however, the larger menu can make the coercer worse off. The comparison identifies what a useful commitment would have to do: preserve punishment after resistance while limiting the operation after compliance. A coercer can gain from restricting an action it would rationally choose later because the restriction changes the bargain offered earlier. This gives a specific form to the power to bind oneself (Schelling 1960).

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Supporting Information

Contents

A. Notation

Let $y = x$ after concession and $y = 0$ after resistance. A subwar operation produces a gain $S(y) = \min\{z, 1 - y\}$ and costs the coercer λc_1, so $S(0) = z$. Under Assumption 1(i), $S(y) = z$ at every relevant accepted demand. Settlement after resistance costs the coercer a, and $h = p + c_2$ is the target's loss from war. As in the main text, $W(y)$ is the probability of war after transfer y, while $L(y)$ is the probability of either war or the operation. When the operation is active, $W(y) = F(\kappa_W(y))$ and $L(y) = F(\kappa_L(y))$. When it is dominated, $W(y) = L(y) = F(\kappa_S(y))$. Finally, $\alpha = 1 - L(x^*)$ is the probability of settlement after compliance and $\pi = W(0)$ is the probability of war after resistance.

B. Proofs of the Lemmas

Lemma 1 (The Coercer's Menu in the Benchmark): Given transfer y, the coercer fights if and only if $c_1 \leq \kappa^\dagger(y) = p - y + a\mathbb{1}[y = 0]$ and settles otherwise.

Lemma 2 (No Type Prefers Resistance): For every c_1 and $x > 0$, the coercer's continuation value is weakly higher after concession than after resistance and strictly higher for types that do not fight after concession. If the operation is active after concession, the indifferent types are $[0, \kappa_W(x)]$, which has positive measure if and only if $p > x + S(x)$. If the operation is dominated, the indifferent types are $[0, \kappa_S(x)]$.

Lemma 3 (The Coercer's Menu with a Subwar Operation): If $\kappa_W(y) < \kappa_L(y)$, the coercer fights below $\kappa_W(y)$, uses the operation between the cutoffs, and settles above $\kappa_L(y)$. If the cutoffs are reversed, the operation is dominated. Equation (3) gives both cutoffs.

Lemma 4 (The Target's Rule): The target concedes x if and only if (4) holds.

B.1. Proof of Lemma 1

Proof. After transfer y, settlement pays $y - a\mathbb{1}[y = 0]$ and war pays $p - c_1$. The coercer fights when $p - c_1 \geq y - a\mathbb{1}[y = 0]$. Rearranging this inequality gives the cutoff in the lemma. $\square$

B.2. Proof of Lemma 2

Proof. Fix c_1 and $x > 0$. Settlement pays x after concession and $-a$ after resistance. War pays $p - c_1$ after either response. The operation pays $x + S(x) - \lambda c_1$ after concession and $z - \lambda c_1$ after resistance. If $z \leq 1 - x$, then $x + S(x) = x + z > z$. If $z > 1 - x$, then $x + S(x) = 1 > z$. Thus each continuation action pays at least as much after concession as after resistance.

War is the only action whose payoff does not strictly rise after concession. Because the war cutoff weakly decreases with y, a type that fights after concession also fights after resistance. That type receives $p - c_1$ after either response and is indifferent.

Now take a type that does not fight after concession. Because war was available but not chosen, its continuation value after concession exceeds $p - c_1$. Settlement and the operation also pay strictly more after concession because $x > -a$ and $x + S(x) > z$. The type's continuation value after resistance is the maximum of $p - c_1$, $-a$, and $z - \lambda c_1$. Each candidate is strictly below its corresponding post-concession payoff or below the post-concession continuation value. The type therefore strictly prefers concession.

If the operation is active, the indifferent set is exactly $[0, \kappa_W(x)]$ and has positive measure if and only if $p > x + S(x)$. If the operation is dominated, the indifferent set is $[0, \kappa_S(x)]$. These types' payoffs do not determine their demands. The pooling equilibrium assigns them x^*.

No type gains by inducing resistance. Settling types prefer the largest accepted demand. Operation users weakly prefer a larger demand and strictly prefer it under Proposition 2's condition that $S(y) = z$. Fighting types are indifferent. The pooling equilibrium assigns the same demand to every type, so the target evaluates that demand under the prior F. $\square$

B.3. Proof of Lemma 3

Proof. Fix transfer y. The coercer's three payoffs are

$$u_S = y - a\mathbb{1}[y=0], \quad u_W = p - c_1, \quad u_L = y + S(y) - \lambda c_1$$

Their slopes in c_1 are 0, -1, and $-\lambda$. Since $-1 < -\lambda < 0$, the optimal action changes from war to the operation and then to settlement as c_1 rises. Pairwise indifference gives the cutoffs in (3): $u_W = u_L$ at κ_W, $u_L = u_S$ at κ_L, and $u_W = u_S$ at κ_S.

The operation is optimal for some types if and only if $\kappa_W < \kappa_L$. If $\kappa_W \geq \kappa_L$, its payoff never exceeds $\max\{u_W, u_S\}$. The coercer then fights below κ_S and settles above it. □

B.4. Proof of Lemma 4

Proof. After concession, war occurs with probability $W(x)$ and gives the target $1 - p - c_2$. The operation occurs with probability $L(x) - W(x)$ and gives the target $1 - x - S(x)$. Settlement occurs otherwise and gives the target $1 - x$. Hence

$$V_2(x) = W(x)(1-h) + (1-x)(1-W(x)) - S(x)(L(x) - W(x))$$

The target's payoff after resistance is $V_2(0)$, evaluated at $x = 0$ and $S(0) = z$. The target concedes if and only if $V_2(x) \geq V_2(0)$. Rearranging yields (4). □

Remark 4 (Distributional Scope): Let $r(c) = \frac{f(c)}{1-F(c)}$ be the upper-tail hazard of a differentiable war-cost distribution. On the uncapped domain, the sufficient condition

$$\sup_{0 \leq c \leq \frac{p}{1-\lambda}} r(c) < \frac{1-\lambda}{h}$$

makes the target's acceptance margin strictly decreasing, so its accepted demands form an interval. For the uniform distribution this condition is exactly Assumption 1(ii). The continuation cutoffs, the target rule, Proposition 3, Proposition 8, and the threat–assurance identity do not require uniformity. The quadratic demands and uniqueness claims that use this interval argument do. Outside the hazard condition, the paper makes no connected-acceptance-set claim without a global solver.

Proof. Let $G(x) = V_2(x) - V_2(0)$ be the target's acceptance margin. In a region where the operation is dominated after compliance,

$$G'(x) = f(p-x)(h-x) - [1-F(p-x)].$$

In an active, uncapped region,

$$G'(x) = f\Big(\kappa_{W(x)}\Big)\frac{h-x-z}{1-\lambda} - \Big[1-F\Big(\kappa_{W(x)}\Big)\Big].$$

If either parenthesized harm term is nonpositive, the corresponding derivative is negative immediately. Otherwise divide by the positive survival probability. Since every relevant cutoff lies below $\frac{p}{1-\lambda}$ and each harm term is at most h, the stated upper-tail-hazard bound makes both derivatives strictly negative. The payoff, and hence G, is continuous where the operation enters or leaves the coercer's envelope, so the accepted set is an interval on the uncapped domain. For $F(c) = \frac{c}{\bar{c}}$, $r(c) = \frac{1}{\bar{c}-c}$; its supremum on the stated interval is $\frac{1}{[\bar{c}-\frac{p}{1-\lambda}]}$. Substitution reduces the bound to $\bar{c} > \frac{p+h}{1-\lambda}$. □

C. Proof of the Equilibrium Characterization

C.1. Proof of Proposition 1 and Proposition 2

Proof. Benchmark. Let $G^\dagger(x)$ be the target's payoff from concession minus its payoff from resistance. With interior cutoffs, the target concedes exactly when

$$x^2 + x(\bar{c} - p - h) - ha \le 0$$

The constant term is negative, so the quadratic has one positive and one negative root. The target accepts every nonnegative demand up to the positive root and rejects every larger demand. The positive root is (1).

At this demand, war occurs with probability $F(p+a)$ after resistance and $F\big(\kappa^\dagger(x^\dagger)\big)$ after concession. Definition 1 and the binding acceptance condition then give (2) and $\alpha^\dagger = 1 - F\big(\kappa^\dagger(x^\dagger)\big)$.

Operation game. Let $G(x) = V_2(x) - V_2(0)$. First consider demands for which the operation is dominated after concession. Then $W(x) = L(x) = F(p-x)$, and the uniform distribution gives

$$G'(x) = \frac{h+p-\bar{c}-2x}{\bar{c}} < 0$$

Assumption 1(ii) implies $\bar{c} > h + p$, so this derivative is negative. Next consider demands for which the operation is active after concession and $S(x) = z$. Then $L(x) = F\big(\frac{z}{\lambda}\big)$, and

$$G'(x) = \frac{f(\kappa_W)(h - x - z)}{1 - \lambda} - (1 - F(\kappa_W))$$

This derivative is negative if $\frac{h+p-2x-2z}{1-\lambda} < \bar{c}$. Its left side is no larger than $\frac{h+p}{1-\lambda}$, which is below $\bar{c}$ under the maintained restriction. The target's payoff is continuous where the operation enters the coercer's optimal menu. Thus G is strictly decreasing across the dominated and active regions wherever $S(x) = z$.

As $x \to 0^+$, the audience cost ceases to apply because the target has conceded. The operation–settlement cutoff therefore falls from $\kappa_L(0) = \frac{z+a}{\lambda}$ to $\frac{z}{\lambda}$. Although κ_W does not contain the audience cost, the probability of war can also jump if the operation is dominated on one side of $x = 0$, since κ_S contains a. Combining both changes gives

$$G(0^+) = z[L(0) - L(0^+)] + (h - z)[W(0) - W(0^+)]$$

Both bracketed differences are nonnegative. Removing the audience cost raises the settlement payoff and leaves the war and operation payoffs unchanged, so each type's optimal action after concession of 0^+ is weakly less forceful than after resistance. Hence $L(0) \geq L(0^+)$ and $W(0) \geq W(0^+)$. The coefficient $h - z$ multiplies a nonzero difference only when the operation is dominated at 0^+: if the operation is active on both sides of $x = 0$, then $\kappa_W(0) = \kappa_W(0^+) = \frac{p-z}{1-\lambda}$ and the second bracket vanishes. Domination at 0^+ requires $\kappa_W(0^+) \geq \kappa_L(0^+) = \frac{z}{\lambda}$, which rearranges to $z \leq \lambda p < h$, so the coefficient is positive whenever the difference is. Therefore $G(0^+) \geq z[L(0) - L(0^+)]$, with equality when the operation is active on both sides of $x = 0$. If it is active after both responses and $\kappa_L(0) \geq \bar{c}$, then $L(0) = 1$ and $L(0^+) = F\big(\frac{z}{\lambda}\big)$. Hence $G(0^+) = z\alpha$. Because $z > 0$, a positive demand is accepted if and only if $\alpha > 0$. The audience cost by itself does not determine this result.

Proposition 2 assumes $\alpha > 0$ and $S(x) = z$ at every accepted demand. Demands at which the bound binds are therefore rejected, and strict monotonicity over the remaining interval gives a unique largest accepted demand x^*. This also proves Remark 3. If $z \geq \lambda\bar{c}$, then $\alpha = 0$ and the quadratic's constant term $-(1 - \lambda)\bar{c}z\alpha$ is zero. Its roots are 0 and $-((1 - \lambda)\bar{c} - p +$

$2z - h$). No positive demand is then accepted, so $x^* = 0$ and $\Omega^* = 0$; conversely $\alpha > 0$ gives $G(0^+) = z\alpha > 0$ and a positive accepted demand.

Equilibrium strategies. Lemma 2 establishes that no coercer type prefers resistance. In the operation game, every type demands x^*, the target concedes, and types below $\kappa_W(x^*)$ fight, so war occurs with probability $W(x^*)$. In the benchmark, every type demands $x^\dagger$, and types below $\kappa^\dagger(x^\dagger)$ fight.

Intuitive criterion. Let $U_C(x, c_1)$ be a type's optimized continuation payoff after concession of x, and let $U_R(c_1)$ be its optimized payoff after resistance. In the benchmark these functions are

$$U_C^\dagger(x, c_1) = \max\{x, p - c_1\}, \quad U_R^\dagger(c_1) = \max\{-a, p - c_1\} \tag{12}$$

In the operation game they are

$$U_C(x, c_1) = \max\{x, p - c_1, \min\{x + z, 1\} - \lambda c_1\}, \quad U_R(c_1) = \max\{-a, p - c_1, z - \lambda c_1\} \tag{13}$$

In either game, $U_C(x, c_1)$ is weakly increasing in x. Lemma 2 gives $U_R(c_1) \leq U_C(x^e, c_1)$, where x^e denotes the relevant pooling demand.

Apply the continuous-type version of the intuitive criterion. For an off-path demand x', first remove a type if its equilibrium payoff exceeds the most it can receive under any target response that is optimal for some posterior. The target then places weight only on the remaining types. A candidate fails the criterion only if some remaining type strictly prefers x' under every best response to such beliefs.

First take $x' < x^e$. Monotonicity gives $U_C(x', c_1) \leq U_C(x^e, c_1)$, while Lemma 2 gives $U_R(c_1) \leq U_C(x^e, c_1)$. No target response can therefore make any type strictly better off. The positive-measure interval of types that fight after concession of x^e is not equilibrium dominated: these types also fight after either response to x'. A posterior supported on that interval makes the target indifferent, so the maintained tie rule selects concession, but no type profits.

Now take $x' > x^e$. Suppose first that concession is optimal for the target under some posterior. Because $U_C(x', c_1) \geq U_C(x^e, c_1)$ for every type, no type is equilibrium dominated. The prior is therefore an admissible off-path belief. Under the prior, the target strictly rejects x': x^e is the largest accepted demand, and the target concedes at equality. Rejection gives every type $U_R(c_1) \leq U_C(x^e, c_1)$. If concession is not optimal under any posterior, rejection is optimal under every posterior; a belief supported on the equilibrium fighting types is admissible and again deters the deviation. Thus every off-path demand has an intuitive-criterion-consistent belief and sequentially rational response under which no type gains. Both pooling equilibria satisfy the criterion.

D1. The stronger D1 test also compares types that survive equilibrium dominance. Let $q \in [0,1]$ be the probability that the target concedes after x'. For each type, let $\mathcal{D}_{c_1}$ contain the values of q that make the deviation strictly profitable and let $\mathcal{D}^0_{c_1}$ contain those that make it payoff-equivalent. D1 prunes c_1 if $\mathcal{D}_{c_1} \cup \mathcal{D}^0_{c_1}$ is a proper subset of $\mathcal{D}_{\hat{c}_1}$ for some other type $\hat{c}_1$.

In the benchmark, take $x' = x^\dagger + \varepsilon < p$ and define $k = p - x^\dagger$ and $k' = p - x'$. Types $c_1 \leq k'$ fight after either target response, so $\mathcal{D}_{c_1} = \emptyset$ and $\mathcal{D}^0_{c_1} = [0,1]$; D1 cannot prune them. Types in $(k', k]$ fight on the equilibrium path and after resistance but settle after concession of x', so $\mathcal{D}_{c_1} = (0,1]$ and $\mathcal{D}^0_{c_1} = \{0\}$. Types above k require a strictly positive concession probability and are pruned by the switching types. The D1-admissible support is therefore $[0,k]$, not only the switching interval. The target is indifferent for the always-fighting types and strictly prefers concession for the switching types because $x' < p \leq p + c_2$. Under the maintained tie rule, every D1-admissible belief induces concession, and the switching types profit.

The same calculation applies in the operation game. Choose $x' = x^* + \varepsilon$ so that $x' + z < p$, and define $k = \kappa_W(x^*)$ and $k' = \kappa_W(x')$. Types below k' fight after either response and cannot be pruned. Types in $(k', k]$ fight on the equilibrium path and after resistance but use the operation after concession of x'. Types above k require a strictly positive concession probability and are pruned by the switching types. D1 therefore permits beliefs on $[0,k]$. Concession leaves the target indifferent for the always-fighting types and strictly improves its payoff for the switching types because $x' + z < p \leq p + c_2$. The tie rule again induces concession under every admissible belief, so the switching types profit.

If $c_2 > 0$, neither conclusion depends on the tie rule. In the benchmark choose $x' \in (p, p + c_2)$; in the operation game choose $x' \in (p - z, p + c_2 - z)$. Every type in the corresponding D1 support then switches away from war after concession, and the target strictly prefers concession. If $c_2 = 0$ and rejection is instead allowed at off-path indifference, a posterior on types that fight after either response can support rejection. Thus the pooling equilibria fail D1 under the maintained convention, but D1′s conclusion at $c_2 = 0$ is tie-breaking-sensitive. The paper imposes the intuitive criterion rather than D1. Proposition 9 extends this calculation to every equilibrium with pure target responses.

Closed form in the operation game. In the active, interior region, $W = \frac{p-x-z}{(1-\lambda)\bar{c}}$, $W_0 = \frac{p-z}{(1-\lambda)\bar{c}}$, and $W_0 - W = \frac{x}{(1-\lambda)\bar{c}}$. Moreover, $L = \frac{z}{\lambda\bar{c}} = 1 - \alpha$ is constant in x. If $\kappa_L(0) \geq \bar{c}$, every nonfighting type uses the operation after resistance. Hence $L_0 = 1$ and $L_0 - L = \alpha$. The binding acceptance condition becomes

$$\frac{x((1-\lambda)\bar{c} - p + x + z)}{(1-\lambda)\bar{c}} = \frac{x(h-z)}{(1-\lambda)\bar{c}} + z\alpha$$

Rearranging yields $x^2 + x((1-\lambda)\bar{c} - p + 2z - h) - (1-\lambda)\bar{c}z\alpha = 0$. Its constant term is negative because $\alpha > 0$. The unique positive root is therefore (5). □

D. Proofs of the Comparative Statics

D.1. Proof of Remark 1

Proof. At the largest accepted demand, (4) binds, so $V_2(x^*) = V_2(0)$. Let $W = W(x^*)$, $L = L(x^*)$, $W_0 = W(0)$, and $L_0 = L(0)$. Substituting the target's payoffs yields

$$W(1-h) + (1-x^*)(1-W) - S(x^*)(L-W) = W_0(1-h) + (1-W_0) - z(L_0 - W_0)$$

Cancel common terms and collect the terms in x^*:

$$x^*(1-W) = h(W_0 - W) + z(L_0 - W_0) - S(x^*)(L-W)$$

Separating hW_0 from hW gives (6). Definition 1 gives $\Omega^* = x^*(1-W)$. Therefore coercive ability is zero if and only if $\Theta^* = \Delta^*$. □

D.2. Proof of Proposition 3

Proof. Suppose the operation is active after concession and the operation–settlement cutoff is interior. By Lemma 3, the coercer settles when $c_1 > \kappa_L(x^*)$. Thus $\alpha = 1 - F(\kappa_L(x^*))$.

With a general benefit $S(y)$ and additional cost $C(y)$, settlement pays y, while the operation pays $y + S(y) - \lambda c_1 - C(y)$. The transfer cancels from their difference. The actions are equally valuable at $\kappa_L(y) = \frac{S(y)-C(y)}{\lambda}$, with derivative $\frac{\partial \kappa_L}{\partial y} = \frac{S'(y)-C'(y)}{\lambda}$. In the benchmark, the corresponding war–settlement cutoff is $\kappa^\dagger(y) = p - y$, whose derivative is -1. Each additional unit transferred therefore lowers the cutoff one-for-one and makes settlement more likely.

Under the present specification, Assumption 1(i) makes $S(y) = z$ at every accepted demand, and there is no additional cost, so $C(y) = 0$. Consequently, $\frac{\partial \kappa_L}{\partial y} = 0$, $\kappa_L(x^*) = \frac{z}{\lambda}$, and $\alpha = 1 - F\left(\frac{z}{\lambda}\right)$. Since F is strictly increasing at this cutoff, $\frac{\partial \alpha}{\partial z} = -\frac{f\left(\frac{z}{\lambda}\right)}{\lambda} < 0$ and $\frac{\partial \alpha}{\partial \lambda} = f\left(\frac{z}{\lambda}\right)\frac{z}{\lambda^2} > 0$.

The zero derivative is specific to the fixed-effect specification. If $S(y) = z(1-y)$ or operating cost rises with the transfer, then $\frac{\partial \kappa_L}{\partial y} < 0$. A larger transfer would make settlement more likely, although less rapidly than in the benchmark. $\square$

D.3. Proof of Corollary 1

Proof. Settlement after transfer y pays $y - a\mathbb{1}[y=0]$ and the operation pays $y + g - \lambda c_1$. Their difference is $g + a\mathbb{1}[y=0] - \lambda c_1$, which vanishes at $\kappa_L(y) = \frac{g+a\mathbb{1}[y=0]}{\lambda}$. The target's loss d does not enter the coercer's comparison and therefore does not appear in the cutoff. After compliance $\kappa_L(x) = \frac{g}{\lambda}$ and $\alpha = 1 - F\left(\frac{g}{\lambda}\right)$, which is zero exactly when $g \geq \lambda \bar{c}$.

Now set $g = 0$. With no share conveyed, the operation pays the coercer $y - \lambda c_1$ after transfer y against y from settlement, so every type with $c_1 > 0$ settles after compliance: $\alpha = 1$, $W(x) = L(x) = F(p - x)$, and the corner of Remark 3, which requires the operation–settlement cutoff to reach the top of the type support, is unreachable. After resistance the operation pays $-\lambda c_1$ against $-a$ from settlement and $p - c_1$ from war, so $\kappa_W(0) = \frac{p}{1-\lambda}$ and $\kappa_L(0) = \frac{a}{\lambda}$, and the operation is active if and only if $a > \lambda(p + a)$. Because the games share the post-compliance node, the target's loss from conceding x is the same strictly increasing function in both, so the largest accepted demand rises with expected loss after resistance. With interior post-resistance cutoffs that loss is $\frac{h(p+a)}{\bar{c}}$ in the benchmark and $\frac{hp}{(1-\lambda)\bar{c}} + \frac{d\left(\frac{a}{\lambda} - \frac{p}{1-\lambda}\right)}{\bar{c}}$ with the operation, and their difference has the sign of $(a(1-\lambda) - \lambda p)(d - \lambda h)$, whose first factor is positive exactly when the operation is active after resistance. The subwar capability therefore raises the accepted demand and coercive ability if and only if $d > \lambda h$. $\square$

D.4. Proof of Proposition 6

Proof. Benchmark. Types below $\kappa^\dagger = p - x^\dagger$ fight and receive $p - c_1$. The remaining types settle and receive $x^\dagger$. Under the uniform distribution,

$$V^\dagger = \frac{1}{\bar{c}}\left(\int_0^{\kappa^\dagger} (p-c)\,\mathrm{d}c + \int_{\kappa^\dagger}^{\bar{c}} x^\dagger\,\mathrm{d}c\right) = x^\dagger + \frac{1}{\bar{c}}\kappa^\dagger\left(p - \frac{\kappa^\dagger}{2} - x^\dagger\right)$$

Since $p - x^\dagger = \kappa^\dagger$, the term in parentheses is $\frac{\kappa^\dagger}{2}$. Substitution gives the first line of (10).

Operation game. Types below κ_W fight. Types in $(\kappa_W, \kappa_L]$ use the operation and receive $x^* + z - \lambda c_1$. The remaining types settle. Integrating over the three regions gives

$$V^* = x^* + \frac{1}{\bar{c}}\left(\kappa_W(p - x^* - z) - (1-\lambda)\frac{\kappa_W^2}{2} + z\kappa_L - \lambda\frac{\kappa_L^2}{2}\right)$$

Since $p - x^* - z = (1-\lambda)\kappa_W$, the first two terms reduce to $\frac{(1-\lambda)\kappa_W^2}{2}$. Since $\kappa_L = \frac{z}{\lambda}$, the remaining operation term becomes $\frac{z^2}{2\lambda}$. Substituting $\kappa_W = \frac{p-x^*-z}{1-\lambda}$ yields the second line of (10).

Define Q_0 and Q_1 as in the main text. An interior operation-game war cutoff at x^* implies $p - x^* > z > 0$, while Assumption 1(ii) implies $p - x^* < \bar{c}$. The benchmark cutoff evaluated at x^* is therefore interior. Adding and subtracting $Q_0(x^*)$ gives the main-text decomposition. Simplification gives

$$Q_1(x^*) - Q_0(x^*) = \frac{(\lambda(p - x^*) - z)^2}{2\lambda(1-\lambda)\bar{c}} \geq 0$$

which establishes the nonnegative menu gain. Moreover, $Q_{0'}(y) = 1 - \frac{p-y}{\bar{c}} > 0$ whenever the benchmark cutoff is interior. Thus $Q_0(x^*) - Q_0(x^\dagger) < 0$ if $x^* < x^\dagger$. The equilibrium payoff falls exactly when this loss exceeds the menu gain. □

D.5. Proof of Proposition 7

Proof. On the selected equilibrium path, the target concedes. Total conflict cost is zero under settlement, $c_1 + c_2$ under war, and λc_1 under the operation. The demand x and the operation's gain z do not enter because each is transferred between the states.

Let $b = \kappa^\dagger(x^\dagger) = p - x^\dagger$. If the operation is evaluated after a concession $x^\dagger - d$, its war cutoff is

$$\kappa_W(x^\dagger - d) = \frac{p - x^\dagger + d - z}{1 - \lambda} = \frac{b + d - z}{1 - \lambda}$$

while $\kappa_L(x^\dagger - d) = \frac{z}{\lambda}$ is constant. At $d = d_W$, the war cutoff equals b and the operation remains active because $b < \frac{z}{\lambda}$. Below this threshold, $\kappa_W(x^\dagger - d) < b$ and war falls. Above it, the active-branch war cutoff exceeds b; if the operation later becomes dominated, the war cutoff becomes $p - (x^\dagger - d) = b + d > b$. Hence the subwar capability lowers war exactly when $d < z - \lambda b = d_W$.

While the operation is active, conflict cost after concession $x^\dagger - d$ is

$$K^L(d) = \int_0^{\kappa_W(x^\dagger - d)} (c_1 + c_2)\, dF(c_1) + \lambda \int_{\kappa_W(x^\dagger - d)}^{\kappa_L(x^\dagger - d)} c_1\, dF(c_1)$$

and $\Delta K(d) = K^{L(d)} - K^\dagger$. Differentiating with respect to d gives

$$\frac{\mathrm{d}\Delta K(d)}{\mathrm{d}d} = f\big(\kappa_W(x^\dagger - d)\big)\left[\kappa_W(x^\dagger - d) + \frac{c_2}{1 - \lambda}\right] > 0 \tag{14}$$

because $\frac{\mathrm{d}\kappa_W(x^\dagger - d)}{\mathrm{d}d} = \frac{1}{1-\lambda}$ and the upper operation cutoff is constant.

If the operation becomes dominated, the continuation menu reduces to war and settlement with cutoff $b + d$. Conflict cost is then $\int_0^{b+d} (c_1 + c_2)\, dF(c_1)$, whose derivative in d is $f(b + d)(b + d + c_2) > 0$. Conflict cost is continuous where the operation leaves the menu because all three actions have the same payoff there. Thus $\Delta K(d)$ is strictly increasing throughout the feasible range.

At $d = d_W$, the two war cutoffs coincide: $\kappa_W(x^\dagger - d_W) = b$. Types below b fight in both games, while types between b and $\kappa_L(x^\dagger - d_W)$ replace benchmark settlement with the operation. Hence

$$\Delta K(d_W) = \lambda \int_b^{\kappa_L(x^\dagger - d_W)} c_1\, dF(c_1) > 0 \tag{15}$$

Since $\Delta K(d)$ is continuous and strictly increasing, $\Delta K(0) < 0$ implies a unique root $d_K \in (0, d_W)$. The signs on either side give the three regions stated in the proposition. If $\Delta K(0) \geq 0$, strict monotonicity gives $\Delta K(d) > 0$ for every $d > 0$.

Finally, $b > 0$ and $\lambda > 0$ imply $d_W = z - \lambda b < z$. Thus $d \geq z$ implies $d > d_W$ and therefore $\Delta K(d) > 0$.

It remains to vary the operation's relative cost. Let $q = 1 - \lambda$ and $r(x) = p - x - z$. On an active interior branch with a uniform cost distribution,

$$K(x, \lambda) = \frac{1}{2\overline{c}} \left[r(x) \frac{r(x) + 2c_2}{q} + \frac{z^2}{\lambda} \right].$$

Holding the concession fixed and differentiating gives

$$\frac{\partial K(x, \lambda)}{\partial \lambda} = \frac{1}{2\overline{c}} \left[r(x) \frac{r(x) + 2c_2}{q^2} - \frac{z^2}{\lambda^2} \right].$$

This derivative is negative if and only if (11) holds. When $c_2 = 0$, operation activity requires $\frac{r(x)}{q} < \frac{z}{\lambda}$, which is exactly the strict inequality needed to make the derivative negative.

Finally,

$$\frac{\partial K(x, \lambda)}{\partial x} = -\frac{r(x) + c_2}{q\overline{c}} < 0. \tag{16}$$

Along the equilibrium path,

$$\frac{\partial K^*}{\partial \lambda} = \frac{\partial K(x^*, \lambda)}{\partial \lambda} + \frac{\partial K(x^*, \lambda)}{\partial x} \frac{\partial x^*}{\partial \lambda}.$$

If (11) holds at $x = x^*$ and $\frac{\partial x^*}{\partial \lambda} \geq 0$, the first term is strictly negative and the second is weakly negative. Therefore $\frac{\partial K^*}{\partial \lambda} < 0$, which proves the final claim. □

Proposition 8 (The Post-Compliance Cutoff Is Selection-Free): Consider any equilibrium and any demand conceded with positive probability. If the operation is active after that demand and $S(x) = z$, the coercer settles if and only if $c_1 > \frac{z}{\lambda}$. This cutoff is the same at every such demand and under every off-path belief. If the operation is active after every conceded demand, so every accepted demand exceeds $p - \frac{z}{\lambda}$, then the settling types are $\left(\frac{z}{\lambda}, \overline{c}\right]$ in every equilibrium.

D.6. Proof of Proposition 8

Proof. Fix an equilibrium and a transfer $y = x > 0$ that the target concedes. The coercer then chooses among settlement, war, and the operation, with payoffs x, $p - c_1$, and $x + S(x) - \lambda c_1$. These payoffs depend only on c_1, x, and the model parameters. Because the target has already moved, they do not depend on its beliefs or the demands made by other types.

The operation payoff minus the settlement payoff is $S(x) - \lambda c_1$. When $S(x) = z$, this difference is $z - \lambda c_1$ and does not depend on x. The operation is preferred to settlement exactly when $c_1 < \frac{z}{\lambda}$. Because the operation is active by hypothesis, Lemma 3 implies that types above $\kappa_L(x) = \frac{z}{\lambda}$ settle and $\alpha = 1 - F\left(\frac{z}{\lambda}\right)$.

The cutoff is therefore $\frac{z}{\lambda}$ at every conceded demand for which the operation is active. Activity itself depends on the demand. Equation (3) implies that the operation enters the coercer's envelope after concession of x exactly when $x > p - \frac{z}{\lambda}$. In a separating equilibrium with two accepted demands, the operation may be active after the larger demand and dominated after the smaller one. The corresponding settlement cutoffs would then be $\frac{z}{\lambda}$ and $\kappa_S(x_L)$. The common settlement probability $1 - F\left(\frac{z}{\lambda}\right)$ applies when every accepted demand exceeds $p - \frac{z}{\lambda}$. $\square$

D.7. Proof of The Resistance-Stage War Comparison

Proof. If the operation is active after resistance, the coercer fights when $c_1 \leq \kappa_W(0)$. If it is dominated, the coercer fights when $c_1 \leq \kappa_S(0) = p + a$. After resistance, $S(0) = z$ and the audience cost applies. Thus $\kappa_W(0) = \frac{p-z}{1-\lambda}$ and $\kappa_L(0) = \frac{z+a}{\lambda}$, which gives the two probabilities stated in the main text. The operation is active if and only if $\kappa_W(0) < \kappa_L(0)$, or

$$\lambda(p - z) < (1 - \lambda)(z + a)$$

Rearranging gives $\lambda(p + a) < z + a$. The condition for the operation to lower war is $\kappa_W(0) < \kappa_S(0) = p + a$, which rearranges to the same inequality. Thus the operation is active exactly when it lowers war. Assumption 1(ii) implies $\kappa_W(0) \leq \frac{p}{1-\lambda} < \overline{c}$, so the reduction is strict whenever the operation enters the menu. The operation never raises war after resistance. The extension in which the operation carries its own audience cost is derived in §F.4. The Operation's Audience Cost. □

D.8. Proof of Proposition 4

Proof. Benchmark. With an interior war cutoff, $W^\dagger(x^\dagger) = \frac{p-x^\dagger}{\overline{c}}$. Therefore

$$\Omega^\dagger = x^\dagger(1-W^\dagger) = \frac{x^\dagger(\overline{c}-p+x^\dagger)}{\overline{c}}$$

Equation (1) implies $x^{\dagger 2} = ha - x^\dagger(\overline{c}-p-h)$. Substitute this expression into the numerator:

$$x^\dagger(\overline{c}-p) + x^{\dagger 2} = x^\dagger(\overline{c}-p) + ha - x^\dagger(\overline{c}-p-h) = h(x^\dagger + a)$$

Dividing by $\overline{c}$ gives $\Omega^\dagger = \frac{h(x^\dagger+a)}{\overline{c}}$.

Operation game. In the active, interior region, $W(x^*) = \frac{p-x^*-z}{(1-\lambda)\overline{c}}$. Hence $\Omega^* = \frac{x^*((1-\lambda)\overline{c}-p+x^*+z)}{(1-\lambda)\overline{c}}$. Equation (5) gives $x^{*2} = (1-\lambda)\overline{c}z\alpha - x^*((1-\lambda)\overline{c}-p+2z-h)$. Substituting yields

$$\begin{aligned}
&x^*((1-\lambda)\overline{c}-p+z) + x^{*2} \\
&\quad = x^*((1-\lambda)\overline{c}-p+z) + (1-\lambda)\overline{c}z\alpha \\
&\qquad - x^*((1-\lambda)\overline{c}-p+2z-h) \\
&\quad = x^*(h-z) + (1-\lambda)\overline{c}z\alpha
\end{aligned}$$

Divide by $(1-\lambda)\overline{c}$ to obtain $\Omega^* = \frac{x^*(h-z)}{(1-\lambda)\overline{c}} + z\alpha$. Comparing this expression with the benchmark gives (8). □

D.9. Proof of Proposition 5

Proof. Define $\varphi(t) = t^2 + [(1-\lambda)\overline{c}-p+2z-h]t - (1-\lambda)\overline{c}z\alpha$. Its unique positive root is x^*. Assumption 1(ii) implies $(1-\lambda)\overline{c} > p+h$, so $\varphi(-z) = z(p+h-(1-\lambda)\overline{c}-z-(1-\lambda)\overline{c}\alpha) < 0$. Since $x^\dagger - z \geq -z$, this argument cannot lie below the negative root. Therefore $x^\dagger > x^* + z$ if and only if $\varphi(x^\dagger - z) > 0$.

Expand $\varphi(x^\dagger - z)$ and use $x^{\dagger 2} = ha - (\overline{c}-p-h)x^\dagger$ from (1):

$$\begin{aligned}\varphi\big(x^\dagger - z\big) = ha + z^2 - [(1-\lambda)\overline{c} - p + 2z - h]z \\ -(1-\lambda)\overline{c}z\alpha \\ +x^\dagger[(1-\lambda)\overline{c} - p + 2z - h \\ -(\overline{c} - p - h) - 2z]\end{aligned}$$

The coefficient on $x^\dagger$ simplifies to $-\lambda\overline{c}$. Collecting the remaining terms gives

$$\varphi\big(x^\dagger - z\big) = ha + z(p + h - (1-\lambda)\overline{c} - z - (1-\lambda)\overline{c}\alpha) - \lambda\overline{c}\,x^\dagger$$

Positivity of this expression, written in primitives, is

$$(p + c_2)a + z(2p + c_2 - (1-\lambda)\overline{c} - z - (1-\lambda)\overline{c}\alpha) > \lambda\overline{c}\,x^\dagger$$

Dividing $\varphi\big(x^\dagger - z\big) > 0$ by $(1-\lambda)\overline{c}$ and substituting $1 - W\big(x^\dagger - z\big) = \frac{(1-\lambda)\overline{c} - p + x^\dagger}{(1-\lambda)\overline{c}}$ gives (9).

Finally, compare each coercer type's payoff. The same condition gives $x^\dagger > x^* + z$, so for every $c_1 \geq 0$,

$$\max\{x^*, p - c_1, x^* + z - \lambda c_1\} \leq \max\big\{x^\dagger, p - c_1\big\}$$

The inequality is strict when $p - c_1 < x^\dagger$. These are precisely the types that settle after compliance in the benchmark. They have positive measure because the benchmark cutoff is interior.

War after compliance. By (3) and Lemma 1, $\kappa_W(x^*) > \kappa^\dagger\big(x^\dagger\big)$ is equivalent to $\frac{p - x^* - z}{1-\lambda} > p - x^\dagger$ and hence to $\big(x^\dagger - x^* - z\big) + \lambda\big(p - x^\dagger\big) > 0$. The first addend is strictly positive because $x^\dagger > x^* + z$ and the second is strictly positive because the benchmark post-compliance war cutoff is interior. Both cutoffs lie strictly inside $[0, \overline{c}]$ under the stated hypotheses, so F is strictly increasing between them and $W(x^*) = F(\kappa_W(x^*)) > F\big(\kappa^\dagger\big(x^\dagger\big)\big)$. □

D.10. Interior Switch Accounting

Hold transfer y fixed, suppose $S(y) = z$, and let every continuation cutoff be interior. Write $A(y) = a\mathbb{1}[y = 0]$ and define the operation's entry margin

$$M(y) = z + (1-\lambda)A(y) - \lambda(p - y).$$

The operation enters the coercer's menu when $M(y) > 0$. Relative to the benchmark, the masses that switch from war and settlement to the operation are

$$m_{W(y)} = \frac{M(y)}{(1-\lambda)\overline{c}}, \quad m_{S(y)} = \frac{M(y)}{\lambda\overline{c}}.$$

The first equality follows from $\kappa_{S(y)} - \kappa_{W(y)} = \frac{M(y)}{1-\lambda}$ and the second from $\kappa_{L(y)} - \kappa_{S(y)} = \frac{M(y)}{\lambda}$.

A former fighter lowers the target's loss by $h - z$, while a former settler raises it by z. The change in expected harm at history y is therefore

$$zm_{S(y)} - (h-z)m_{W(y)} = \frac{M(y)(z-\lambda h)}{\lambda(1-\lambda)\overline{c}}. \tag{17}$$

This expression has the sign of $z - \lambda h$. Applying it after both target responses and using $M(0) - M(x) = (1-\lambda)a - \lambda x$ shows that the change in threat relative to the change in assurance has the sign of $(z-\lambda h)((1-\lambda)a - \lambda x)$. The first factor determines whether the operation raises harm at a given history, and the second determines whether more types switch after resistance or compliance.

The closed-form region truncates the settlement-to-operation switch after resistance because $\kappa_{L(0)} \geq \overline{c}$. There, $\Theta^* < \Theta^\dagger$ and $\Delta^* > \Delta^\dagger$ hold together exactly when

$$(1-\lambda)(\overline{c} - p - a)z < M(0)(h-z), \quad \frac{M(x^*)(z-\lambda h)}{\lambda(1-\lambda)} + h\big(x^\dagger - x^*\big) > 0. \tag{18}$$

The first inequality says that war replacement reduces punishment after resistance despite the operation's effect on former settlers. The second says that post-compliance harm rises once the smaller accepted demand is included. This is an exact conditional statement, not a claim about prevalence in a parameter population.

E. Separating Demands and D1

Lemma 5 (Separating Demands Do Not Change the Selected Outcome): Consider either game under the conditions of Proposition 1 or Proposition 2 and any perfect Bayesian equilibrium in which the target's response is a pure function of the demand. Let $\overline{x}$ be the largest demand the target accepts and let x^e denote $x^\dagger$ in the benchmark and x^* in the operation game. Every type that does not fight after concession of $\overline{x}$ demands $\overline{x}$. Only types that fight after both target responses may choose another demand, and their separation does not change the target's acceptance condition at $\overline{x}$. Hence $\overline{x} \leq x^e$. The pooling equilibrium attains x^e and is therefore coercer-optimal even when separating demands are allowed.

Proof. Conceding zero weakly improves the target's payoff, so the accepted set is nonempty. Lemma 2 and Assumption 1(i) imply that each type's payoff after concession is weakly increasing in the demand and strictly increasing for any type that does not fight afterward. Assumption 1(ii) leaves a positive measure of such types at every accepted demand. If the accepted set had no maximum, these types would have no optimal demand. It therefore has a largest element $\overline{x}$, and every type that does not fight after its concession must demand it.

A type that fights after concession of $\overline{x}$ also fights after resistance by Lemma 2. Such a type gives the target payoff $1 - h$ after either response, so it contributes zero to the target's payoff difference between concession and resistance. All other types demand $\overline{x}$. Bayes' rule therefore implies that the sign of the target's payoff difference at $\overline{x}$ is the same under its on-path posterior as under the prior: separating always-fighting types changes only terms whose difference is zero. Thus any largest demand accepted in a separating equilibrium is also accepted under the prior and cannot exceed the largest prior-accepted demand x^e. Propositions 1 and 2 show that the pooling equilibrium attains x^e. Because every type's continuation payoff is weakly increasing in an accepted demand, no separating equilibrium makes any type better off. $\square$

The D1 calculation in the proof of Proposition 1 and Proposition 2 is local to the two pooling equilibria. The same comparison applies to every equilibrium in the class the paper compares, and its verdict is the same in both games.

Proposition 9 (No Accepted Demand Survives D1 Under Pure Target Responses): Consider either game under Assumption 1 and the rule that the target concedes when indifferent. Take any perfect Bayesian equilibrium in which the target's response is a pure function of the demand, and let $\overline{x}$ be the largest demand the target accepts. Every coercer type's equilibrium payoff is then $U_C(\overline{x}, c_1)$, so the equilibrium with the largest accepted demand is the one every type weakly prefers within this class. If the war cutoff after concession of $\overline{x}$ is interior, the equilibrium fails D1. If $c_2 > 0$, the failure does not depend on the tie rule.

E.1. Proof of Proposition 9

Proof. Write $\kappa(x)$ for the war cutoff after concession of x: it equals $\kappa_W(x)$ where the operation is active after that demand, $\kappa_S(x) = p - x$ where the operation is dominated, and $\kappa^\dagger(x) = p - x$ in the benchmark. On an uncapped local interval the branch expressions agree where the operation enters or leaves the envelope, so κ is continuous and strictly decreasing there. Lemma 2 gives $\kappa(x) \leq \kappa(0)$, so a type that fights after concession also fights after resistance. Assumption 1(ii) gives $\kappa(x) \leq \frac{p}{1-\lambda} < \overline{c}$, so at every demand a positive measure of types does not fight after concession.

The largest accepted demand and the equilibrium payoff. Conceding 0 raises each type's settlement payoff from $-a$ to 0 and leaves its war and operation payoffs unchanged, so each type's optimal action after concession of 0 is weakly less forceful than after resistance and the target's payoff from conceding 0 is weakly higher than from resisting. Under the tie rule the target concedes 0, so the accepted set is nonempty. Lemma 2 and the monotonicity of $U_C(x, c_1)$ in x imply that every type weakly prefers any accepted demand to any rejected demand, strictly so for types that do not fight after concession. Those types have positive measure, and their payoff is strictly increasing in the accepted demand and strictly above their payoff from any rejected demand, so they would have no optimal demand if the accepted set had no maximum. The accepted set therefore attains a largest element $\overline{x}$, which they demand. Optimality then gives $u^*(c_1) = U_C(\overline{x}, c_1)$ for every type: for types pooled at $\overline{x}$ by construction; for a type at a smaller accepted demand because monotonicity and optimality force equality; and for a type at a rejected demand because optimality gives $U_R(c_1) \geq U_C(\overline{x}, c_1)$ while Lemma 2 gives the reverse.

Pruning at an upward deviation. Let $k = \kappa(\overline{x})$ and suppose $0 < k < \overline{c}$. An interior k implies $\overline{x} + z < p$: if the operation is active after $\overline{x}$, then $k > 0$ is $p - \overline{x} - z > 0$; if it is dominated, then $\kappa_W(\overline{x}) \geq \kappa_L(\overline{x}) = \frac{z}{\lambda}$ gives $z \leq \lambda(p - \overline{x}) < p - \overline{x}$. Choose $\varepsilon > 0$ small enough that $x' = \overline{x} + \varepsilon$ satisfies $x' + z < p$ and $k' = \kappa(x') > 0$. The argument establishing $S(x) = z$ above then makes κ continuous and strictly decreasing from $\overline{x}$ to x', so $k' < k$. Let $q \in [0, 1]$ be the probability that the target concedes after x'. Type c_1's payoff from the deviation is $qU_C(x', c_1) + (1 - q)U_R(c_1)$, so the deviation is strictly profitable exactly when $q\Delta(c_1) > E(c_1)$, where

$$\Delta(c_1) = U_C(x', c_1) - U_R(c_1), \quad E(c_1) = U_C(\overline{x}, c_1) - U_R(c_1)$$

Monotonicity gives $\Delta(c_1) \geq E(c_1) \geq 0$, and Lemma 2 gives $E(c_1) = 0$ exactly for $c_1 \leq k$ and $\Delta(c_1) = 0$ exactly for $c_1 \leq k'$. Three groups follow. Types $c_1 \leq k'$ fight after either response to x', so $\mathcal{D}_{c_1} = \emptyset$ and $\mathcal{D}^0_{c_1} = [0, 1]$. Types in $(k', k]$ have $\mathcal{D}_{c_1} = (0, 1]$ and $\mathcal{D}^0_{c_1} = \{0\}$. Types above k have $E(c_1) > 0$, so $\mathcal{D}_{c_1} \cup \mathcal{D}^0_{c_1} = \left[\frac{E(c_1)}{\Delta(c_1)}, 1\right]$ with $\frac{E(c_1)}{\Delta(c_1)} > 0$, a proper subset of $(0, 1]$. The first two groups have $\mathcal{D}_{c_1} \cup \mathcal{D}^0_{c_1} = [0, 1]$, which no $\mathcal{D}_{\hat{c}_1}$ can properly contain, so D1 cannot prune them; the third group is pruned by the second, which has measure $\frac{k - k'}{\overline{c}} > 0$. Every D1-admissible belief is therefore supported on $[0, k]$.

The target's reply. Under any belief on $[0, k]$, types in $[0, k']$ fight after concession of x' and after resistance, contributing $1 - h$ to both. Types in $(k', k]$ fight after resistance but not after concession of x'; they take at most $x' + z < p \leq h$, so each raises the target's payoff from conceding above $1 - h$. Conceding is therefore weakly better than resisting, and strictly better when the belief places mass on $(k', k]$. Under the tie rule the target concedes.

Conclusion. If x' is off the equilibrium path, D1 confines the belief to $[0, k]$, the target concedes, and every type in $(k', k]$ receives $U_C(x', c_1) > U_R(c_1) = u^*(c_1)$, contradicting equilibrium. If x' is on the path, it is rejected because $\overline{x}$ is the largest accepted demand; every type that plays it must satisfy $U_R(c_1) = U_C(\overline{x}, c_1)$, hence $E(c_1) = 0$ and $c_1 \leq k$, so the belief obtained from Bayes' rule is also supported on $[0, k]$. The computation above then makes conceding weakly better, and the tie rule requires concession, contradicting rejection. Either way the equilibrium fails D1.

Independence of the tie rule. Suppose $c_2 > 0$ and take $x' \in (p - z, p + c_2 - z)$ in the operation game and $x' \in (p, p + c_2)$ in the benchmark. Since $\overline{x} + z < p$, such an x' exceeds $\overline{x}$. In the operation game $x' > p - z$ gives $\lambda(p - x') < \lambda z < z$, so the operation is active after concession of x' and $\kappa(x') = \kappa_W(x') < 0$; in the benchmark $x' > p$ gives $\kappa^\dagger(x') < 0$. No admissible type then fights after concession of x', every admissible type takes at most $x' + z < h$, and the target strictly prefers conceding under every belief supported on $[0, k]$, whether that belief comes from D1 or from Bayes' rule. The switching types again profit.

The excluded boundary. If $k \leq 0$, no type fights after concession of $\overline{x}$, the group $(k', k]$ is empty, and D1 retains only the types that minimize E. The argument does not apply, and a pure-response equilibrium can survive: if $p + a \geq \overline{c}$ in the benchmark, every type fights after resistance, the target accepts exactly the demands $x \leq h$, and at $\overline{x} = h$ the D1-admissible belief is concentrated on $c_1 = 0$, against which the target strictly refuses every larger demand. Propositions 1 and 2 exclude this boundary by assuming interior war cutoffs. □

F. Extensions and Scope

F.1. Escalation and Recontestation

The following remark shows how escalation risk changes the operation–settlement boundary.

Remark 5 (Escalation Risk Partially Restores Assurance): Suppose the operation triggers general war with probability $q \in [0, 1)$, and that general war reopens the dispute as it does elsewhere in the model. The coercer pays λc_1 whenever it uses the operation and pays c_1 in addition when escalation occurs, so after transfer y the operation pays $(1-q)(y+S(y)) + qp - \lambda c_1 - qc_1$ against $y - a\mathbb{1}[y=0]$ from settlement. If the operation is active after compliance and $S(y) = z$, the operation–settlement cutoff is (19). Its derivative in the transfer is $\frac{\partial \kappa_L^q}{\partial y} = -\frac{q}{\lambda+q} < 0$. Its derivative in the escalation probability, evaluated at $q = 0$, is $\frac{\partial \kappa_L^q}{\partial q} = \frac{\lambda(p-y-z)-z}{\lambda^2}$, which is negative wherever the operation is active after compliance because activity requires $z > \lambda(p - y)$. Setting $q = 0$ recovers $\kappa_L(y) = \frac{z}{\lambda}$.

$$\kappa_L^q(y) = \frac{(1-q)z + q(p-y)}{\lambda + q} \tag{19}$$

The remark lets the operation trigger general war with probability q. A later section lets general war reopen only a share of an accepted concession. Either change alters the target's acceptance constraint. The first is signed only locally; the second is solved in closed form. We make no claim that the baseline ordering survives arbitrary escalation or recontestation technologies.

F.2. Post-Compliance Military Advantage

The baseline makes general war equally attractive after either target response. Now leave the resistance node unchanged but suppose that compliance creates a positional advantage in a subsequent war. War after compliance gives payoffs

$$(p + \varepsilon - c_1, 1 - p - \varepsilon - c_2), \quad \varepsilon > 0,$$

while ordinary war after resistance still gives $(p - c_1, 1 - p - c_2)$. The perturbation preserves total war surplus and changes no operation or settlement payoff. Let $h = p + c_2$, $D = (1-\lambda)\bar{c}$, and $\alpha = 1 - \frac{z}{\lambda \bar{c}}$.

Suppose ε is small enough that the largest accepted demands remain positive and all cutoffs stay in the interior regions used by Propositions 1 and 2. In the benchmark, the post-compliance war cutoff is

$$t_\varepsilon^\dagger = p + \varepsilon - x_\varepsilon^\dagger.$$

In the game with the operation, the active post-compliance cutoffs are

$$\kappa_{W,\varepsilon}^* = \frac{p + \varepsilon - x_\varepsilon^* - z}{1 - \lambda}, \quad \kappa_L^* = \frac{z}{\lambda}.$$

The resistance cutoffs are unchanged in both games. The two accepted demands are the positive roots of

$$\left(x_\varepsilon^\dagger\right)^2 + (\overline{c} - 2p - c_2 - 2\varepsilon)x_\varepsilon^\dagger + \varepsilon(2p + c_2 + \varepsilon) - ha = 0, \tag{20}$$

$$(x_\varepsilon^*)^2 + (D - 2p - c_2 + 2z - 2\varepsilon)x_\varepsilon^* + \varepsilon(2p + c_2 + \varepsilon - 2z) - Dz\alpha = 0. \tag{21}$$

Proposition 10 (Post-Compliance Advantage Weakens Coercion): Suppose the target and coercer use pure strategies, $z < h + \varepsilon$, and the accepted demands lie in the maintained full-effect, interior region. Each game has a pooling PBE at the demand given by (20) or (21). Every pure-strategy PBE with an accepted positive demand has the same outcome. A larger ε strictly lowers the accepted demand and coercive ability and strictly raises war and expected total conflict cost. As $\varepsilon \to 0$, the equilibrium outcome converges to its baseline value.

The coercive-ability expressions are

$$\Omega_\varepsilon^\dagger = x_\varepsilon^\dagger \left(1 - \frac{t_\varepsilon^\dagger}{\overline{c}}\right), \quad \Omega_\varepsilon^* = x_\varepsilon^* \left(1 - \frac{\kappa_{W,\varepsilon}^*}{\overline{c}}\right). \tag{22}$$

Expected total conflict costs are

$$K_\varepsilon^\dagger = \frac{\left(t_\varepsilon^\dagger\right)^2 + 2c_2 t_\varepsilon^\dagger}{2\overline{c}}, \quad K_\varepsilon^* = \frac{(1-\lambda)\left(\kappa_{W,\varepsilon}^*\right)^2 + 2c_2\kappa_{W,\varepsilon}^* + \frac{z^2}{\lambda}}{2\overline{c}}. \tag{23}$$

Proof. We first construct the pooling equilibria. At either root, the target is indifferent under the prior, so the maintained tie rule selects compliance. Relative to resistance, compliance with a positive demand raises the coercer's payoff from war by ε, from the operation by x, and from settlement by $x + a$. Every type therefore strictly prefers the pooling signal to a resisted signal. In the benchmark, a belief on a sufficiently high-cost type supports resistance after any off-path positive demand. In the operation game, a belief on an operation user supports resistance at a demand at most z, and a belief on a type that uses the operation after resistance but settles after compliance supports resistance above z. At zero, a belief on a type that fights after compliance makes resistance strict because resistance avoids the additional loss ε. These beliefs deter every deviation.

Target indifference in the benchmark is

$$(h+\varepsilon)F\big(t_\varepsilon^\dagger\big) + x_\varepsilon^\dagger\big(1 - F\big(t_\varepsilon^\dagger\big)\big) = hF(p+a).$$

In the operation game it is

$$(h+\varepsilon)W(x_\varepsilon^*) + (x_\varepsilon^* + z)(L(x_\varepsilon^*) - W(x_\varepsilon^*)) + x_\varepsilon^*(1 - L(x_\varepsilon^*)) = hW(0) + z(L(0) - W(0)).$$

Substituting the uniform distribution, $L(0) = 1$, and $L(x_\varepsilon^*) = \frac{z}{\lambda \bar{c}}$ gives (20) and (21).

We next show that pure separation cannot change a positive-compliance outcome. Consider any pure-strategy PBE with an accepted positive demand. At every accepted demand, the interior conditions leave a positive measure of high-cost types that do not fight after compliance. Their payoff rises strictly with the demand, so they would have no optimal demand if the accepted set had no maximum. Let $\overline{x}$ denote that maximum. For either game, let $U_C^g(x, c_1)$ and $U_R^g(c_1)$ be the coercer's optimized payoffs after compliance and resistance. The action-by-action payoff gains from compliance established above imply

$$U_C^g(x, c_1) > U_R^g(c_1) \quad \text{for every } x > 0.$$

No type can therefore remain at a resisted demand. Moreover, $U_C^g(x, c_1)$ is weakly increasing in x. A type assigned to an accepted $x < \overline{x}$ must consequently satisfy $U_C^g(x, c_1) = U_C^g(\overline{x}, c_1)$. In the maintained full-effect region, the settlement and operation payoffs rise strictly with x. Equality is possible only if the type fights after compliance with both demands.

The target's loss from compliance with such a type is $h + \varepsilon$. After resistance, its loss is h if the type fights, z if it conducts the operation, and zero if it settles. The condition $z < h + \varepsilon$

makes each resistance loss strictly smaller than the compliance loss. Bayes' rule therefore makes the target resist any lower demand chosen by a positive measure of these indifferent types. Almost every type must pool at $\overline{x}$; assignments of prior-null types cannot change any outcome. The posterior at $\overline{x}$ is consequently the prior, so $\overline{x}$ equals the appropriate positive root above.

The same pooling equilibria satisfy the intuitive criterion. A lower off-path demand cannot raise any type's payoff even if accepted. At a higher demand, no type is equilibrium dominated, so the prior remains an admissible belief. The target rejects because the pooling demand is the largest demand accepted under the prior, and rejection leaves every type strictly below its equilibrium payoff. Thus no deviation is profitable under an admissible response.

Let $R_\varepsilon^\dagger = 2x_\varepsilon^\dagger + \overline{c} - 2p - c_2 - 2\varepsilon$ and $R_\varepsilon^* = 2x_\varepsilon^* + D - 2p - c_2 + 2z - 2\varepsilon$. Each is positive because it is the slope of its quadratic at the positive root. Implicit differentiation gives

$$\frac{\partial x_\varepsilon^\dagger}{\partial \varepsilon} = -\frac{h + p + 2\varepsilon - 2x_\varepsilon^\dagger}{R_\varepsilon^\dagger} < 0, \quad \frac{\partial x_\varepsilon^*}{\partial \varepsilon} = -\frac{h + p + 2\varepsilon - 2z - 2x_\varepsilon^*}{R_\varepsilon^*} < 0.$$

The first numerator is the sum of $p + \varepsilon - x_\varepsilon^\dagger$ and $h + \varepsilon - x_\varepsilon^\dagger$; the second is the sum of the corresponding two margins net of z. Interior post-compliance war makes both positive. Each war cutoff therefore rises at rate $\left(1 - \partial \frac{x_\varepsilon}{\partial} \varepsilon\right)$ divided by its cutoff's scale. Equation (22) then strictly falls because both the demand falls and war rises. Differentiating (23) with respect to its war cutoff establishes the cost result.

Finally, both roots and all displayed outcome formulas are continuous at $\varepsilon = 0$. This completes the proof. □

The threat–assurance accounting clarifies the result. The extension changes neither game's punishment after resistance:

$$\Theta_\varepsilon^\dagger = hF(p + a), \quad \Theta_\varepsilon^* = hW(0) + z(L(0) - W(0)).$$

It raises assurance failure directly by making each post-compliance war more damaging and indirectly by lowering the accepted demand, which expands the set of types that fight:

$$\Delta_\varepsilon^\dagger = (h + \varepsilon)F\big(t_\varepsilon^\dagger\big), \quad \Delta_\varepsilon^* = (h + \varepsilon)W(x_\varepsilon^*) + z(L(x_\varepsilon^*) - W(x_\varepsilon^*)). \tag{24}$$

Thus $\Omega_\varepsilon = \Theta_\varepsilon - \Delta_\varepsilon$ in each game. The exact comparison between the games retains the main paper's form, $\left(\Theta^*_\varepsilon - \Theta^\dagger_\varepsilon\right) - \left(\Delta^*_\varepsilon - \Delta^\dagger_\varepsilon\right)$, but its thresholds move with ε. Any strict comparison at a fixed baseline parameter vector persists for a sufficiently small advantage by continuity.

Remark 6 (Military Power with Weaker Assurance): In the same interior regions,

$$\frac{\partial x^\dagger_\varepsilon}{\partial p} = \frac{2x^\dagger_\varepsilon + a - 2\varepsilon}{R^\dagger_\varepsilon}, \quad \frac{\partial x^*_\varepsilon}{\partial p} = 2\frac{x^*_\varepsilon - \varepsilon}{R^*_\varepsilon}.$$

Hence the benchmark demand still rises with military power when $\varepsilon < \frac{a}{2}$. With the operation, it rises if and only if $x^*_\varepsilon > \varepsilon$. If $x^*_\varepsilon \leq \varepsilon$, greater power also strictly lowers coercive ability. The post-compliance advantage can therefore reverse, not merely attenuate, the effect of power when the operation already leaves little room for an acceptable demand.

A fully separating all-rejected PBE also survives. Assign each type a distinct demand above $\max\{p + \varepsilon, h, z\}$ and let the target resist after inferring the type. War is then dominated by settlement after compliance, and the transfer alone exceeds the harm resistance can produce, so resistance is sequentially rational. Every type receives its resistance continuation payoff and is indifferent among rejected signals. This equilibrium has no compliance. Proposition 10 characterizes the positive-compliance class selected in the paper.

Proposition 11 (Backfire with a Post-Compliance Military Advantage): Suppose Assumption 1 and the interior conditions hold. If $x^\dagger_\varepsilon > x^*_\varepsilon + z$, the operation produces strictly less peaceful extraction and gives the coercer a strictly lower ex ante payoff than the benchmark. It also produces weakly more war and weakly greater total conflict cost. War is strictly higher if a positive measure of benchmark settlers fights with the operation; conflict cost is strictly higher if a positive measure fights or conducts the operation.

The condition is the post-compliance-advantage version of the full-gain condition in Proposition 5. Both games now give the coercer the same military advantage after compliance. The loss in the concession exceeds the operation's gain, so the added option cannot compensate for that loss.

Proof. The proof compares continuation outcomes type by type. Write $x_A = x_\varepsilon^\dagger$ and $x_M = x_\varepsilon^*$. After compliance, the optimized coercer payoffs are

$$U_C^\dagger(x_A, c_1) = \max\{p + \varepsilon - c_1, x_A\}$$

and

$$U_C^*(x_M, c_1) = \max\{p + \varepsilon - c_1, x_M + z - \lambda c_1, x_M\}.$$

If $x_A > x_M + z$, each nonwar operation-game branch is below x_A, while the two games share the same war branch. Hence $U_C^*(x_M, c_1) \leq U_C^\dagger(x_A, c_1)$ for every type. The inequality is strict for every type that settles after benchmark compliance, a positive-measure set under the interior cutoff. Integrating proves the strict ex ante payoff comparison.

The same comparison ranks peaceful extraction. A type that fights after benchmark compliance satisfies $p + \varepsilon - c_1 \geq x_A > x_M + z$, so it also fights with the operation and contributes zero in both games. A benchmark settler contributes x_A, while its operation-game contribution is either zero after war or $x_M < x_A$ after settlement or the operation. Peaceful extraction is therefore strictly lower.

Finally, every benchmark fighter remains a fighter with the operation. Benchmark settlers generate no conflict cost, while their operation-game continuation action has nonnegative cost. War and total conflict cost are therefore weakly higher. The stated positive-measure conditions give strictness. This establishes every comparison in the proposition. $\square$

The pure-strategy equilibrium correspondence changes continuously in outcomes but not in form at $\varepsilon = 0$. In the baseline, always-fighting types can separate because war gives them the same payoff after either target response. Every $\varepsilon > 0$ makes acceptance strictly better for those types. They therefore join the accepted pool, and the selected pooling outcome converges to the baseline as their payoff advantage vanishes.

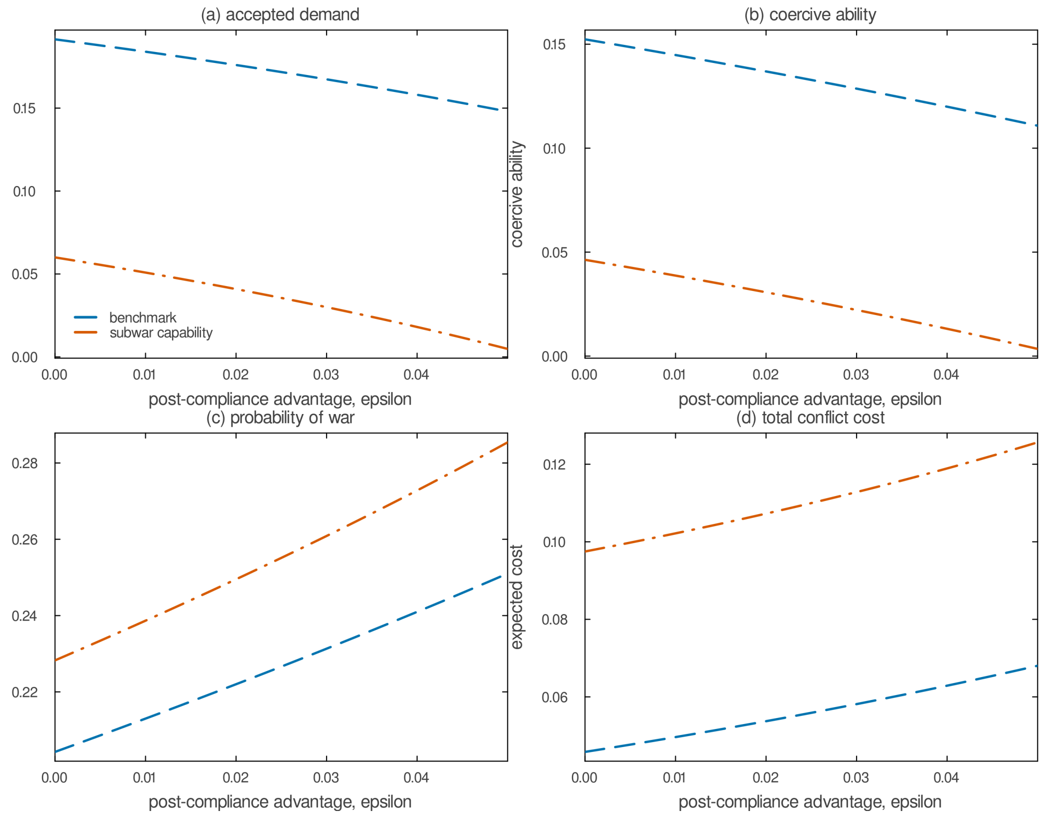

Figure 6: A Post-Compliance Military Advantage Weakens Coercion

Note: Each panel follows the pure-strategy pooling equilibrium as compliance raises the coercer's probability of victory by ε. The dashed blue curve is the benchmark; the dash-dotted orange curve is the game with the subwar capability. Punishment after resistance is fixed along each curve. The accepted demand and coercive ability fall because assurance worsens, while post-compliance war and total conflict cost rise. Parameters: $p = 0.60$, $c_2 = 0.02$, $a = 0.30$, $\lambda = 0.08$, $z = 0.12$, and c_1 uniform on $[0, 2]$.

F.3. Appropriative War

The baseline lets general war reopen the whole dispute after either target response, so an accepted transfer is worth nothing to the coercer once war occurs. This section replaces that assumption with its opposite pole and every point between. Let general war after compliance reclaim only a share of the concession and contest the remainder of the good. Nothing has been handed over after resistance, so that node is unchanged and the extension operates on the post-compliance branch alone.

Remark 7 (Appropriative War): Let $\tau \in [p, 1]$ index how far an accepted transfer deters war after compliance: war then pays the coercer $p + (1-\tau)x - c_1$ and the target $1 - h - (1-\tau)x$. If war reclaims a share ρ of the transfer and contests the rest of the good, then $\tau = \rho + (1-\rho)p$. The baseline is $\tau = 1$; war reclaims nothing at $\tau = p$, where every post-compliance action leaves the transfer in place. After compliance, each war cutoff is the baseline's with τx in place of x,

$$\kappa^{\dagger}(x) = p - \tau x, \quad \kappa_W(x) = \frac{p - \tau x - z}{1 - \lambda}$$

while $\kappa_L(x) = \frac{z}{\lambda}$ and $\alpha = 1 - F\left(\frac{z}{\lambda}\right)$ are unchanged, as is the entire post-resistance node. The target's expected payment is $x(1 - \tau W(x))$ and still equals $\Theta^* - \Delta^*$ for Θ^* and Δ^* as defined in (6). The largest accepted demands are the positive roots of (25), and

$$\Omega^{\dagger} = \frac{h\left(\tau x^{\dagger} + a\right)}{\bar{c}}, \quad \Omega^* = \frac{\tau x^*(h - z)}{(1-\lambda)\bar{c}} + z\alpha$$

Setting $\tau = 1$ recovers (1), (5), and Proposition 4.

$$\begin{gathered} \tau^2 x^2 + (\bar{c} - \tau(p+h))x - ha = 0, \\ \tau^2 x^2 + ((1-\lambda)\bar{c} - \tau(p+h) + 2\tau z)x - (1-\lambda)\bar{c}z\alpha = 0 \end{gathered} \tag{25}$$

The cutoffs follow from the payoffs. After compliance, war pays $p + (1-\tau)x - c_1$, settlement pays x, and the operation pays $x + z - \lambda c_1$, so war beats settlement when $c_1 < p - \tau x$ and beats the operation when $(1-\lambda)c_1 < p - \tau x - z$. Neither settlement nor the operation involves war, so their comparison is the baseline's and Proposition 3 applies unchanged: the transfer cancels, κ_L is constant in the demand, and compliance still buys no restraint on the operation margin. What the transfer buys is war avoidance, and τ is the rate at which it does so. The target's loss from complying is $x(1 - W(x)) + (h + (1-\tau)x)W(x) + z(L(x) - W(x))$, which equals $x(1 - \tau W(x)) + hW(x) + z(L(x) - W(x))$. Equating this to the unchanged loss from resisting and substituting $W(0) - W(x) = \tau x/((1-\lambda)\bar{c})$ gives the second quadratic; setting $z = 0$ and $W =$

$L = F(\kappa^{\dagger})$ gives the first. The Assumption 1(ii) is unchanged and remains sufficient, because it binds most tightly at $\tau = 1$.

Two comparisons change. The demand enters coercive ability only through $\tau x^{\dagger}$ and τx^{*}, so the demand response that drives the result is damped but not removed. War after compliance also responds less to the demand, because the post-compliance war cutoff now moves by τ rather than by one for each unit conceded.

Remark 8 (Payoffs and War Under Appropriative War): Fix $\tau \in [p, 1]$ and suppose the premises of Proposition 5 hold with the cutoffs of the preceding remark. Every coercer type is weakly worse off with the operation if and only if (26) holds, and war after compliance is strictly more likely if and only if (27) holds. When $\tau < 1$ the second condition is the stronger of the two. When $\tau < 1$ the coercer's war payoff after compliance rises in the transfer, so types that fight in both games lose $(1-\tau)(x^{\dagger} - x^{*})$ and the payoff loss is strict for every type rather than for a positive measure. At $\tau = 1$ those types are indifferent and the two conditions coincide, which is Proposition 5.

$$x^{\dagger} + \lambda\kappa^{\dagger}(x^{\dagger}) \geq x^{*} + z \tag{26}$$

$$\tau(x^{\dagger} - x^{*}) + \lambda\kappa^{\dagger}(x^{\dagger}) > z \tag{27}$$

Both conditions come from the same comparison. A benchmark settler receives $x^{\dagger}$; in the operation game it receives at most $\max\{x^{*}, x^{*} + z - \lambda c_1\}$ once c_1 exceeds $\kappa^{\dagger}(x^{\dagger})$, and the operation branch is largest at that cutoff, which gives (26). Below the cutoff the benchmark type fights, and the same comparison delivers the same inequality, so (26) is necessary as well as sufficient. Rearranging $\kappa_W(x^{*}) > \kappa^{\dagger}(x^{\dagger})$ gives (27). The condition stated in Proposition 5, $x^{\dagger} > x^{*} + z$, implies both at $\tau = 1$ but neither is implied by it in reverse: the slack term $\lambda\kappa^{\dagger}(x^{\dagger})$ is the concession a benchmark settler at the war margin already forgoes, and the proposition's condition sets it aside. Proposition 5 is therefore stated on a sufficient condition, not a necessary one, at every τ.

F.4. The Operation's Audience Cost

The main text states the two activity boundaries for an operation that carries its own audience cost. The following remark and derivation are the extension behind that statement.

Remark 9 (An Audience Cost on the Operation): Suppose using the operation imposes an audience cost $a_L \in [0, a]$ after either target response, in addition to the operating cost λc_1. After resistance, $\kappa_L(0) = \frac{z+a-a_L}{\lambda}$ and $\kappa_W(0) = \frac{p-z+a_L}{1-\lambda}$, so the operation is active, and lowers the probability of war, if and only if $z + a - a_L > \lambda(p + a)$. After compliance, the cutoffs on the active branch are $\kappa_L(x) = \frac{z-a_L}{\lambda}$ and $\kappa_W(x) = \frac{p-x-z+a_L}{1-\lambda}$, so the operation is active if and only if $a_L < z - \lambda(p - x)$; on that branch $\alpha = 1 - F\big(\frac{z-a_L}{\lambda}\big)$, and once the inequality fails, the operation is dominated after compliance and $\alpha = 1 - F(p - x)$ instead. On the active closed-form branch, which requires $a_L \leq z + a - \lambda\bar{c}$ to maintain $L(0) = 1$, the binding acceptance condition is (28), and x^* and Ω^* are strictly increasing in a_L when $z < h$. Setting $a_L = 0$ recovers the baseline.

$$x^2 + x((1-\lambda)\bar{c} - p + 2z - a_L - h) - (1-\lambda)\bar{c}z\alpha = 0 \tag{28}$$

The cutoffs follow from the payoffs. Settlement after resistance still pays $-a$, while the operation pays $z - \lambda c_1 - a_L$, which gives the two post-resistance cutoffs; the condition $\kappa_W(0) < \kappa_L(0)$ reduces to $\lambda(p + a) < z + a - a_L$, and comparing $\kappa_W(0)$ with $\kappa_S(0)$ gives the same inequality, so entry and war reduction again coincide. After compliance the transfer is in place and no audience cost attaches to settlement: settlement pays x, the operation pays $x + z - \lambda c_1 - a_L$, and war pays $p - c_1$, which gives the post-compliance cutoffs and both branches of α.

In the active closed-form region, $L(x) = \frac{z-a_L}{\lambda\bar{c}}$ and $L(0) = 1$, while $W(0) - W(x) = \frac{x}{(1-\lambda)\bar{c}}$ because a_L shifts both war cutoffs equally. The substitution used for Proposition 4 gives $\Omega^* = \frac{x^*(h-z)}{(1-\lambda)\bar{c}} + z\alpha$ at the positive root of (28). Implicit differentiation gives

$$\frac{\partial x^*}{\partial a_L} = \frac{x^* + (1-\lambda)\frac{z}{\lambda}}{2x^* + (1-\lambda)\bar{c} - p + 2z - a_L - h} > 0$$

so Ω^* rises with a_L on this branch when $z < h$. Once the operation is dominated after compliance, neither (28) nor its formula for α applies; the comparison follows the war–settlement branch and no global ordering follows from the active-branch derivative.

F.5. The Closed-Form Region

Beyond Assumption 1, Proposition 2′s closed form requires

$$\lambda(p - x^*) < z < p - x^*, \quad \lambda\bar{c} - a \leq z < \lambda\bar{c}, \quad z \leq \frac{1 - p - c_2}{2}$$

The first pair makes the operation active after compliance and keeps the post-compliance war cutoff interior. The second pair combines $\kappa_L(0) \geq \bar{c}$ with $\alpha > 0$ and implies

$$\alpha \leq \min\left\{1, \frac{a}{\lambda\bar{c}}\right\}$$

so the probability of settlement after compliance is capped strictly below one whenever $a < \lambda\bar{c}$. The condition $\kappa_L(0) \geq \bar{c}$ also sets $L(0) = 1$: no coercer type settles after resistance, so the post-resistance menu is effectively binary while the post-compliance menu is ternary. Every closed-form result therefore describes a region in which backing down never occurs, which is the asymmetry that makes Θ^* large and the truncation in (18) one-sided.